QUALIFICATION STANDARD AND PLAN

RQ-S1 SONAR Op

SONAR Operator (SONAR Op) (00324)

NQUAL – AKYF

MITE IDs –
Module 1 – 122046
Module 2 – 122047

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
1 (1.0)		12 Sep 97	
2 (1.1)		6 Jan 00	
3 (1.2)		13 Feb 02	
P-2 (1.3)		20 Sep 06	
P-3 (1.4)		2 Feb 07	
P-4 (1.5)		15 Jun 09	
P-1.4 (1.6)		5 Nov 10	
P-1.5 (1.7)		23 May 12	
1.0 (1.8)		13 Jun 13	CFNOS
1.1 (1.8)		23 Jul 13	CFNOS
1.2 (1.9)		7 Nov 16	CNPTG
1.3 (2.0)	Major	5 Jul 19	NTDC(A)
1.3 (2.0)	Major	9 Aug 19	CNPTG
1.4 (2.1)	Minor	17 Nov 20	NTDC(A)
3.0	Major	30 Nov 21	NTDC(A)
3.0	Major	8 Jan 22	CNPTG

TABLE OF CONTENTS

RECORD OF AMENDMENTS ..... **I**

TABLE OF CONTENTS ..... **II**

ACRONYMS AND ABBREVIATIONS ..... **V**

FOREWARD ..... **IX**

CHAPTER 1: TRAINING MANAGEMENT DETAILS **1**

AIM 2
COURSE DURATION 2
REQUIRED COURSE STAFFING 4
COURSE CAPACITY 4
PREREQUISITES 4
PRIOR LEARNING ASSESSMENT AND RECOGNITION 5
SPECIAL INSTRUCTIONS 5

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE **6**

PO 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES 7
EO 001.01 – Operate Ancillary Acoustic Equipment 8
LG 001.01.01 – Create and upload ematt run geometry 8
LG 001.02.01 – Upload to EMATT 9
PC 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES 10

PO 002 – ANALYZE PASSIVE ACOUSTIC DATA 13
EO 002.01 – Perform PAL 14
LG 002.01.01 – Determining Blade/Shaft Configuration 14
LG 002.01.02 – Identify Aural Sources 16
LG 002.01.03 – Perform Passive Aural Listening 18
EC 002.01 – Perform PAL 19
EO 002.02 – Perform Basic Passive Acoustic Analysis 21
LG 002.02.01 – Classify Weapons Systems 21
LG 002.02.02 – Classify Weapons Systems Lab 22
LG 002.02.03 – Classify Surface Vessels 23
LG 002.02.04 – Classify Surface vessels lab 25
LG 002.02.05 – Classify Diesel Submarines 26
LG 002.02.06 – Classify Diesel Submarines Lab 27
LG 002.02.07 – Classify Nuclear Submarines 28
LG 002.02.08 – Classify Nuclear Submarines Lab 29
LG 002.02.09 – Perform Basic Passive Acoustic Analysis LABs 30
EC 002.02 – Perform Basic Passive Acoustic Analysis 31
LG 002.02.10 – Perform Team Basic Passive Acoustic Analysis 33
PC 002 – ANALYZE PASSIVE ACOUSTIC DATA 35

PO 003 – OPERATE COMMAND SYSTEM 37

EO 003.01 – Record Acoustic Intelligence Data 38
LG 003.01.01 – Operate SPS Recorder 38
EC 003.01 – Record Acoustic Intelligence Data 39
EO 003.02 – Operate Passive Acoustic Equipment 41
LG 003.02.01 – Operate SPS 41
EC 003.02.01 – Operate Passive Acoustic Equipment 43
LG 003.02.02 – Operate SPS for Torpedo Tracking Procedures 46
EC 003.02.02 – Operate SPS for torpedo tracking procedures 47
LG 003.02.03 – Operate SPS for ASW Mission 49
PC 003 – OPERATE COMMAND SYSTEM (SPS) 50

PO 004 – CONDUCT TARGET MOTION ANALYSIS (TMA) 53
EO 004.01 – Operate CMS for TMA 54
LG 004.01.01 – Operate CMS for TMA 54
EC 004.01 – Operate CMS for tma 55
EO 004.02 – Validate TMA Data 58
LG 004.02.01 – Line of Sound 58
LG 004.02.02 – Doppler Theory & Frequency Relationships 59
LG 004.02.03 – Range Calculations & Relative Velocity 61
LG 004.02.04 – Geographic Strip Plot 62
EC 004.02 – Validate TMA Data 63
PC 004 – CONDUCT TMA 66

PO 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION 70
EO 005.01 – Interpret Intelligence/Oceanographic Data 71
LG 005.01.01 – Interpret Oceanographic Products 71
LG 005.01.02 – Sound Propagation 72
LG 005.01.03 – Effects of Water Masses on ASW Sensors 73
LG 005.01.04 – Factors Affecting ASW Ops in Coastal Waters 75
LG 005.01.05 – FOM Calculations 76
EC 005.01 – Interpret Intelligence/Oceanographic Data 77
EO 005.02 – Determine Oceanographic Effects on Acoustic Operations 80
LG 005.02.01 – Setting up oceanographic equipment 80
EC 005.02.01 – Setting Up / Operating Oceanographic Equipment 81
LG 005.02.02 – Calculating acoustic sensor range information 84
LG 005.02.03 – ASW Mission Brief 85
LG 005.02.04 – Create Oceanographic Brief 86
EC 005.02.02 – Determine Oceanographic Effects on Acoustic Operations 87
PC 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION 90

CHAPTER 3: ASSESSMENT OF TRAINEE 93

GENERAL 94
ENABLING AND PERFORMANCE CHECKS 94
RESPONSIBILITY FOR TESTING 94
PROGRESS MONITORING 94
TRAINING REVIEW BOARD 95
COURSE REPORTS 95

ANNEX A – REFERENCES 1

ANNEX B – TASK LIST 1

ANNEX C – RECORD OF DECISIONS 1

ANNEX D – TASK ANALYSIS DIAGRAM 1

ANNEX E – COSTING INFORMATION 1

ANNEX F – ASSESSMENT PLAN 1

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES 1

ANNEX H – SCHEDULING INFORMATION 1

ANNEX I – RECOMMENDED SCHEDULE 1

MODULE 1 SCHEDULE 1
MODULE 2 SCHEDULE 9

ACRONYMS AND ABBREVIATIONS

A	
ACDR	Active Counter Detection Range
ADAC	Acoustic Data Analysis Centre
ADR	Acoustic Data Recorder
ADR	Active Detection Range
AOP	Area Of Probability
A/P	Active / Passive
ARP	Acoustic Range Prediction
ARPS	Acoustic Range Prediction System
ARS	Acoustic Response Sheet
ASW	Anti-Submarine Warfare
ATP	Allied Tactical Publication
ATT	Actual Training Time
AXP	Allied Experimental Publication
B	
BAT	Base Administrative Time
BRG	Bearing
BT	Bathythermograph
C	
CAF	Canadian Armed Forces
CANTASS	Canadian Towed Array Sonar System
CCS	Command & Control System
CFCD	Canadian Forces Controlled Document
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CMS	Combat Management System
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
CPA	Closest Point of Approach
CSNI	Consolidated Secret Network Infrastructure
CTC	Contact
D	
DAOD	Defence Administrative Orders and Directives
dB	Decibel
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DS	Directing Staff
DTG	Date Time Group
DWAN	Defence Wide Area Network
E	
EC	Enabling Check
EDI	Explain Demonstrate Imitate
EMATT	Expendable Mobile ASW Training Target
EMCON	Emissions Control

EO	Enabling Objective
EXTAC	Experimental Tactics
F	
FVEY	Five Eyes
FOM	Figure Of Merit
G	
GIOPS	Global Ice-Ocean Prediction System
GPNet	General Purpose Network
H	
HALT	High Away Low Towards
HMCS	Her Majesty's Canadian Ship
HMS	Hull Mounted Sonar
HOD	Head of Department
I	
IAW	In Accordance With
ID	Identification
IMS	Information Management Standard
IOT	In Order To
IS	Information Systems
IT	Information Technology
L	
LG	Lesson Guidance
LO	Learning Objective
LOCEDGGWY	N2R Edge Node software
LOS	Line Of Sound
LOFAR	Low Frequency Analyser and Recorder
M	
MAD	Magnetic Acoustic Detection
MARTI	Maritime Tactical Instruction
METOC	Meteorological and Oceanographic Centre
MFW	Multi-Function Workstation
MILOC	Military Oceanography
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
MS	Master Sailor
N	
N2R	NITES2R
NAVORD	Naval Order
NCPM	Naval Combat Procedure Manual
NDSOD	National Defence Security Orders and Directives
NFS	Naval Fleet School
NITES2R	Naval Integrated Tactical Environmental System II Redesign
NOAA	National Oceanic and Atmospheric Administration
NOSTP	Naval Operations School Training Publication
NQual	National Qualification
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order

NTDC(A)	Naval Training Development Centre (Atlantic)
NWTP	Naval Warfare Training Publication
O	
OAA	Ownship Aspect Angle
OCS	Ownship Course & Speed
OFA	Ocean Features Analysis
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
OS	Operating System
OSA	Ownship Speed Across
OSI	Ownship Speed Into
P	
PAL	Passive Aural Listening
PC	Performance Check
PLAR	Prior Learning Assessment and Recognition
PO	Performance Objective
POSSUB	Possible Submarine
PT	Physical Training
PTP	Portable Target Programmer
Q	
QA	Quality Assurance
QSP	Qualification Standard and Plan
R	
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
Rel Vel	Relative Velocity
RF	Radio Frequency
RPM	Revolutions Per Minute
RSA	Resultant Speed Across
RSI	Resultant Speed Into
S	
S1	Sailor First Class
SAR	Search And Rescue
SAT	Satisfactory
SCS	Sonar Control Supervisor
SEACOT	Synthetic Environment Advanced Combat Operations Trainer
SEAWOLF	Synthetic Environment Advanced Warfare and Operations Leadership Facility
SLC	Sonobuoy Launch Container
SO	Standing Order
SOP	Standard Operating Procedure
SPS	Sonobuoy Processing System
SPSO	Sonobuoy Processing System Operator
SRC	Speed Related Component
SRPM	Shaft Revolutions Per Minute
SST	Sea Surface Temperature
STM	System Track Management
SVP	Sound Velocity Profile

T	
TAC	Tactical Acoustic Column
TAS	Towed Array Ship
TAT	Training Administrative Time
TA	Training Authority
TAA	Targets Aspect Angle
TBO	Target's Bearing to Ownship
TCS	Target's Course & Speed
TD	Temporary Duty
TDA	Tactical Decision Aid
TE	Training Establishment
TEL	Technology Enabled Learning
TG	Task Group
TGT	Target
TMA	Target Motion Analysis
TO	Training Officer
TPO	Training Petty Officer
TRB	Training Review Board
TS	Training Site
TSA	Target Speed Across
TSI	Target Speed Into
TTT	Training Travel Time
U	
UNSAT	Unsatisfactory
USN	United States Navy
UWW	Under Water Warfare
UWWD	Under Water Warfare Director
V	
VMW	Virtual Machine Ware
Vol	Volume
W	
WHMIS	Workplace Hazardous Material Information Sheet
WIN RGA	Windows RGA
X	
XBT	Expendable Bathythermograph
XSV	Expendable Sound Velocity

FOREWARD

1. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: **http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page**:

   a. A-PD-055/PP-002/PP-002 – SONAR OP MOS ID 00324 – Dated 2018-03-20.

2. A trainee will be granted the National Qualification (NQual) AKYF when the standards listed in all POs have been met.

3. A trainee will be granted the following Military Individual Training and Education (MITE) course identification (ID) when standards listed in all POs have been met:

   a. Module 1: POs 002, 004, 005 – MITE 122046; and

   b. Module 2: POs 001, 003 – MITE 122047.

4. The designated TEs for this training include:

   a. Module 1: POs 002, 004, 005 – all NFSs within NPTG; and

   b. Module 2: POs 001, 003 – ADAC IAW NAVORD 4500-15.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this QSP is to establish the requirements and training for the RQ-S1 SONAR Op (AKYF). This QSP is designed to assist NTDC(A) in developing Module (Mod) 1 and ADAC in developing Mod 2, NFSs and ADAC in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation	
Module 1 – Common SONAR Ops Acoustic Operator	**Periods**
Sum of all EOs/POs	196
Enabling Checks (ECs)	36
Performance Checks (PCs)	19
Actual Training Time (ATT) Sub-total	**251 periods**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22 Annex B Table 3.1	
Base Administrative Time (BAT)	10
Training Administrative Time (TAT)	10
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	24
Sub-total	44
Total Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **8 (50 min) periods** *** Partial days must be rounded up**	**295 periods** **37 days**

Course Duration Calculation	
Module 2 – Fleet SONAR Ops Acoustic Operator Only	**Periods**
Sum of all EOs/POs	118
Enabling Checks (ECs)	18
Performance Checks (PCs)	12
Actual Training Time (ATT) Sub-total	**142 periods 18 days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22 Annex B Table 3.1	
Base Administrative Time (BAT)	2
Training Administrative Time (TAT)	2
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	12
Sub-total	16
Total Course Duration Calculation (ATT+BAT+TTT+TAT+PT) 8 (50 min) periods * Partial days must be rounded up	**158 periods 20 days**

REQUIRED COURSE STAFFING

3. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Course Officer	Lt(N)	1	ALBV	Level II SECRET	All
Senior Instructor	PO1	1	ALBV	Level II SECRET	All
Instructor	PO2	1	ALBD	Level II SECRET	005
Instructor	MS or PO2	1	AEJR	Level II SECRET	002
*Instructor	MS or PO2	1	AEJR	Level II SECRET	002.01.03
Go-Operator	MS	1	ALBC	Level II SECRET	003.02.02
Instructor	MS	1	ALBC	Level II SECRET	001, 003, 004, 005

4. If running at maximum load IAW para 5, instructors with an "*" are required.

COURSE CAPACITY

5. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

a. Maximum – 12;

b. Minimum – 4; and

c. Optimum – 8.

PREREQUISITES

6. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Possess, at minimum, a Level II security clearance;

b. Must have successfully completed the following:

(1) Junior SONAR Operator – RQ-S2 SONAR Op (ALFU); or
(2) CANSUBFOR Approved Waiver; with
(3) Indication via email or message from TRINITY ADAC on the successful completion of 6 x Acoustic Analysis Training Packages; and

c. IAW NPI 13 Annex C.

PRIOR LEARNING ASSESSMENT AND RECOGNITION

7. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

SPECIAL INSTRUCTIONS

8. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES

1. **Performance**: Conduct Preparations for Launching Expendables.

2. **Conditions**:

   a. Given:

      (1) Safety Equipment,
      (2) References,
      (3) Stand-Alone computer,
      (4) EMATT w/ canister, and
      (5) Portable Target Programmer (PTP),

   b. Denied: Nil; and

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at sea.

3. **Standard**: The acoustic operator shall conduct preparations for launching expendables by:

   a. Operating Ancillary acoustic equipment (C0025) by:

      (1) Creating run geometry, and
      (2) Uploading run geometry into PTP, and

   b. Preparing underwater expendable for launch (C0050).

4. **Training Limitations**: Nil.

EO 001.01 – OPERATE ANCILLARY ACOUSTIC EQUIPMENT
LG 001.01.01 – CREATE AND UPLOAD EMATT RUN GEOMETRY

LG 001.01.01 – Create and Upload EMATT Run Geometry		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		
Time	**Classification**	**Instructor**
350 minutes	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Create run geometry to include; a. Acoustic profile b. Broadband profile c. Run profile d. Evasion control parameters e. Evasion profile 2) Upload run geometry to PTP		1) EDI
Scenario	Nil.	
Activity Steps	1) Instructor will explain and demonstrate the process of creating run geometry in WINRGA to include: (40 minutes – Method 1) a. Acoustic profile b. Broadband profile c. Run profile d. Evasion control parameters e. Evasion profile 2) Trainees will then practice creating run geometry (one per piece of equipment) using the EC/PC checklist as guidance on criteria that must be included. (200 minutes – Method 1) 3) Instructor will demonstrate proper procedures to upload run geometry to the PTP. (10 minutes – Method 1) 4) Trainees will then practice uploading run geometry to PTP. (100 minutes – Method 1)	
Instructor Resources		**References**
1) Stand Alone laptop with Windows XP or above OS 2) WINRGA program 3) PTP 4) PC 001 checklist		1) NCPM 600 2) Mk 39 (EMATT) Operators Manual 3) Mk 39 (EMATT) Field Programmability User Guide 4) WHMIS MSDS Sheets
Note(s) to Instructor		
1) Ensure proper sequence of cables connections is followed. 2) PTP must be fully charged the day prior to teaching lesson. 3) Verify all equipment functional prior to instructing lesson.		

CO NTDC(A) Approval - 30/11/2021
CNPTG Approval - 08/01/2022
Version No: 3.0

EO 001.02 – PREPARE UNDERWATER EXPENDABLE FOR LAUNCH
LG 001.02.01 – UPLOAD TO EMATT

LG 001.02.01 – Upload to EMATT		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		After LG 001.01.01
Time	**Classification**	**Instructor**
450 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Upload run profile to EMATT 2) Program EMATT 3) Perform pre-launch checks		1) EDI 2) Quiz
Scenario	Nil.	
Activity Steps	1) Instructor to explain and demonstrate the process for uploading run profiles to the EMATT.(20 min – Method 1) 2) Trainees are to then practice uploading run profiles with instructor support. Each trainee to practice a minimum of 2 uploads. 10 min per trainee/per upload plus reset time and debrief. (280 min – Method 1) 3) Any trainees who are not hands on with the gear during the practice uploads, are to individually create a 10 question quiz on PO 001 content, with an answer key. (Method 2) 4) Instructor to then explain and demonstrate the pre-launch preparations. (10 min – Method 1) 5) Trainees are then to take turns preparing the EMATT for launch. (140 min – Method 1) 6) Any trainees who are not hands on with equipment during the pre-launch preps, are to be given one of the other trainees prepared quiz's to complete while awaiting their turn. (Method 2)	
Instructor Resources		**References**
1) Stand Alone Laptop with Windows XP or above OS 2) WINRGA program 3) PTP 4) EMATT 5) PC 001 checklist		1) NCPM 600 2) Mk 39 (EMATT) Operators Manual 3) Mk 39 (EMATT) Field Programmability User Guide 4) WHMIS MSDS Sheets
Note(s) to Instructor		
Nil.		

PO 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES
PC 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES

PC 001 – Conduct Preparations for Launching Expendables		
Scope	**Location Conditions**	**Instructional Sequence**
PO 001	Unclassified Classroom	After LG 001.02.01
Time	**Classification**	**Instructor**
300 min	UNCLASS	RQ-MS SONAR OP

Scenario	IAW PC 001 Scenario
Assessment Guidance	**Input**: Trainee will be in a classroom provided a standalone laptop with WINRGA program, PTP and EMATT. **Performance:** Trainee must individually use the WINRGA program to create a run geometry, upload it to the PTP and download it to the EMATT IAW PC 001 checklist. (20 mins per trainee) **Output:** Trainee must achieve 33/41 SAT to pass this PC. **Feedback:** Immediate verbal and written feedback IAW PC 001 checklist. (5 mins per trainee)

Instructor Resources	References
1) Stand Alone Laptop with Windows XP or above OS 2) WINRGA program 3) PTP 4) EMATT 5) PC Scenario 6) PC 001 checklist	1) NCPM 600 2) Mk 39 (EMATT) Operators Manual 3) Mk 39 (EMATT) Field Programmability User Guide 4) WHMIS MSDS Sheets

Note(s) to Instructor
Nil.

Assessment Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

PC 001 CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES			
Item	**SAT**	**UNSAT**	**Remarks**
Input Run Geometry			
Create acoustic profile to include:			
Frequency Source	☐	☐	
Amplitude (dB) level	☐	☐	
Smooth transition	☐	☐	
Speed affected	☐	☐	
Set speed dependant parameters	☐	☐	
Create a broadband profile to include:			
Center frequency	☐	☐	
Bandwidth	☐	☐	
Amplitude	☐	☐	
Create a run profile to include:			
Depth change	☐	☐	
Course change	☐	☐	
Speed change	☐	☐	
A/P Sel	☐	☐	
Seg Dur	☐	☐	
Range	☐	☐	
Tonal attn	☐	☐	
AEM	☐	☐	
MAD	☐	☐	
Set evasion control parameters to include:			
Trigger Count (with description)	☐	☐	
Trigger Window	☐	☐	
Lockout time (with description)	☐	☐	
Create an evasion profile to include:			
Depth change	☐	☐	
Course change	☐	☐	
Speed change	☐	☐	
A/P Select	☐	☐	
Segment Duration	☐	☐	
Range	☐	☐	
Tonal attenuation	☐	☐	
AEM	☐	☐	

MAD	☐	☐	
Select independent pinger mode	☐	☐	
Save run profile	☐	☐	
Program EMATT			
Confirm battery powered up	☐	☐	
Connect laptop to PTP	☐	☐	
Download saved run geometry to PTP	☐	☐	
Verify run geometry to PTP	☐	☐	
Connect PTP to EMATT	☐	☐	
Transfer run geometry to EMATT	☐	☐	
Prepare to Launch EMATT			
Remove SLC from over pack	☐	☐	
Remove breakout cap	☐	☐	
Select run select switch to desired run	☐	☐	
Reload target tail first back into SLC	☐	☐	

Score: /41

The trainee must achieve 33/41 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – ANALYZE PASSIVE ACOUSTIC DATA

1. **Performance**: Analyze Passive Acoustic Data (C0065).

2. **Conditions**:

   a. Given:

      (1) Audible Acoustic Data,
      (2) Chronoscope/Stopwatch,
      (3) Computer with Passive Acoustic Analysis Software,
      (4) Headphones,
      (5) Calculator,
      (6) References,
      (7) Logs, and
      (8) Supervision,

   b. Denied: Nil; and

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at sea.

3. **Standard**: The acoustic operator shall analyze passive acoustic data (C0065) by:

   a. Performing Passive Aural Listening (PAL) (C0195); and

   b. Performing basic passive acoustic analysis (F0035) to include:

      (1) Analyzing LOFAR Grams (D0080), and
      (2) Classifying Passive Acoustic Data (C0080).

4. **Training Limitations**: Nil.

EO 002.01 – PERFORM PAL
LG 002.01.01 – DETERMINING BLADE/SHAFT CONFIGURATION

LG 002.01.01 – Determining Blade/Shaft Configuration		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		
Time	**Classification**	**Instructor**
100 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Determine shaft speed (SRPM) to include; a. RPM counting b. Measurement over 6/10 sec c. Converting RPM to hertz d. Aural listening techniques 2) Blade shaft configuration to include; a. blade rate b. shaft rate c. number of blades d. RPM changes 3) Number of shafts		1) Lecture 2) EDI: Assisted aural listening Unassisted aural listening
Scenario	Nil.	
Activity Steps	1) Instructor to conduct PowerPoint lecture covering the following: a. Determine shaft speed (SRPM) to include: (Method 1) i.RPM counting ii.Measurement over 6/10 sec iii.Converting RPM to hertz iv.Aural listening techniques b. Blade shaft configuration to include: i.blade rate ii.shaft rate iii.number of blades iv.RPM changes v.Number of shafts 2) Embedded within the lecture are example sound wave files where instructor will conduct EDI, providing trainee's opportunity to conduct PAL with assistance. Trainees will conduct assisted aural listening and unassisted aural listening. Trainees will utilize the acoustic worksheets for each example sound wave file. (100 min – Method 2)		
Instructor Resources		**References**
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers/headphones		1) CFCD 113 Volumes 1 & 2 2) NAVORD 3470-1 Maritime Acoustic Security Policy

LG 002.01.01 – Determining Blade/Shaft Configuration	
5) Acoustic worksheets 6) Stopwatches	
Note(s) to Instructor	
1) All held blade and shaft configurations must be covered.	

EO 002.01 – PEFORM PAL
LG 002.01.02 – IDENTIFY AURAL SOURCES

LG 002.01.02 – Identify Aural Sources		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.01.02
Time	**Classification**	**Instructor**
300 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Determine source characteristics 2) Identify man-made sources 3) Classify man-made sources 4) Identify environmental sources 5) Classify environmental sources 6) Classification of contacts aurally		1) Lecture 2) EDI
Scenario	Nil.	
Activity Steps	1) Instructor to conduct PowerPoint lecture covering all LOs. Embedded within the lecture are example sound profiles where instructor will conduct EDI, providing trainee's opportunity to conduct first level PAL with assistance. Trainees will utilize the acoustic worksheets for each example sound profile. (300 min – Method 1 & 2) 2) Examples must include but not limited to: a. Torpedoes b. Environmental noise c. Compressed cavitation d. Transients (e.g. Own ship noise) e. Shaft rub f. Gun fire g. Blade resonance h. Water shot/discharge i. Singing screw j. Aircraft k. Blade flutter l. Frying fat m. In/out n. Babbling brook o. Exhaust flutter p. Foghorn q. Diesel audible r. Baby cry s. Whine t. Trawl u. Blade slap v. Buoy	

LG 002.01.02 – Identify Aural Sources	
Instructor Resources	**References**
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers 5) Acoustic worksheets	1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor	
1) Any new sources must be covered in detail as this will be their first exposure to new aural characteristics. 2) Mention recording platform (own ship noise isn't included in CFCD 113 – draft change 2021).	

EO 002.01 – PERFORM PAL
LG 002.01.03 – PERFORM PASSIVE AURAL LISTENING

LG 002.01.03 – Perform Passive Aural Listening		
Location Conditions	**Instructional Sequence**	
SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.01.02	
Time	**Classification**	**Instructor**
700 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Conduct PAL 2) Classify acoustic sources	1) Practical (Scenario Based Learning) 2) EDI	
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various aural example where they must utilize previous learned skills IOT classify acoustic sources. 2) Trainees will be provided 70 contacts total, where there will be a reduction in assistance as they progress in skill. (10 contacts per period for a total of 14 periods of practical – Method 1). 3) Each lab consists of 10 contacts, followed by 30 min allotted debrief time. After each 50 min period, instructor to take up the activity, providing feedback and Directing Staff (DS) solution for the 10 contacts conducted. 4) Throughout trainee work period, instructor to walk throughout and provide coaching with assistance/feedback.	
Instructor Resources	**References**	
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Audio files 6) Headsets for trainees 7) Chronoscope/stopwatch 8) Calculator 9) Instructor Directing Staff solution	1) CFCD 113 Volumes 1 & 2	
Note(s) to Instructor		
1) Historically this has been a challenging skill, anticipate there may be a great deal of coaching required. 2) Due to complexity, in cases with high trainee (>6) load, second instructor will be required.		

EO 002.01 – PERFORM PAL
EC 002.01 – PERFORM PAL

EC 002.01 – Perform PAL		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01	SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.01.03
Time	**Classification**	**Instructor**
150 min	SECRET FVEY	RQ-MS SONAR OP

Scenario	Nil.
Assessment Guidance	**Input:** Trainees will be provided 10 PAL contacts and all resources listed. **Performance:** Trainees will conduct PAL IOT classify 10 contacts. (100 mins) **Output:** IAW ADAC PAL checklist, trainees must obtain 60% for SAT on EC 002.01 (SRPM must be within 10% tolerance). **Feedback:** Provided immediate verbally and written IAW EC 002.01 checklist upon completion of the EC. (50 mins for debrief)

Instructor Resources	**References**
1) Acoustic worksheets 2) Audio files 3) Headset for trainees 4) Chronoscope/stopwatch 5) Calculator 6) Individual SECRET workstations 7) PAL worksheet	1) CFCD 113 Volumes 1 & 2

Note(s) to Instructor
1) All notes material is classified material and must be stored and destroyed IAW NDSODs.

Assessment Limitations	No assistance provided (unlike on ship where you may have support).
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

UNCLASSIFIED

Contact Number: _____ **Contact Location:** _____ **Measurement Date / Time:** _____

Sensor: _____ **Sensor Depth (ft):** _____ **SLD (ft):** _____ **AN:** _____

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPRM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry... Ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2% tolerance	
Class Ind/ID Swath	
1	1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Drive Mode	RR
D/E	

Shaft and Engine Related Sources

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed
Class Indicators		
1	1	1
Modulation/Banding		
1	1	1
Others		
	1	

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1

Transients, Significant Events, Tactical Info

Time	Details
	1 pt for each *significant* event listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1

1/2 harmonics must be listed

Augmentors 1
for the frequency and calling it an augmentor

Auxiliaries
1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter

Others 1
1pt awarded to other significant sources

Unknowns 0

Aural Characteristics

Time	Details
	1 pt for each significant aural characteristic listed

Classification

4 points total
1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

PROTECTED B (when complete)

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.01 – CLASSIFY WEAPONS SYSTEMS

LG 002.02.01 – Classify Weapons Systems		
Location Conditions	**Instructional Sequence**	
SECRET Classroom (ADAC classroom/mobile classroom)	After EC 002.01	
Time	**Classification**	**Instructor**
50 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Identify blade configuration 2) Identify launch occurrence 3) Identify search and attack modes 4) Classify weapons systems	1) Interactive Lecture 2) EDI	
Scenario	Nil.	
Activity Steps	1) Instructor to conduct lecture to cover LOs. Embedded within the lecture are example sound profiles and LOFAR grams where instructor will conduct EDI, providing trainee's opportunity to conduct first level analysis with assistance. (50 mins – Method 1 & 2)	
Instructor Resources	**References**	
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers 5) LOFAR grams(s)	1) CFCD 113 Volumes 1 & 2	
Note(s) to Instructor		
1) Examples are to include all available current weapons systems. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

CO NTDC(A) Approval - 30/11/2021
CNPTG Approval - 08/01/2022
Version No: 3.0

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.02 – CLASSIFY WEAPONS SYSTEMS LAB

LG 002.02.02 – Classify Weapons Systems Lab		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.01
Time	**Classification**	**Instructor**
300 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Analyze LOFAR grams (weapons systems) 2) Identify weapons systems by passive acoustic data 3) Classify passive acoustic data (weapons systems)		1) Practical (Scenario Based Learning)
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various LOFAR gram examples where they must utilize previous learned skills IOT classify weapons systems. (15 mins) 2) Trainees will be provided 3 labs (10 contacts per lab), where there will be a reduction in assistance as they progress in skill. (1 lab = 80 min per lab). (240 mins total – Method 1) 3) After each lab is complete, instructor to take up the activity, providing feedback and Directing Staff solution for the contacts conducted. Throughout trainee labs, instructor to walk throughout and provide coaching with assistance/feedback (15 min per lab). (45 mins total)	
Instructor Resources		**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets for trainees 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Historically this has been a challenging skill, anticipate that coaching will be required. 2) Examples are to include all available current weapons systems. 1 instructor per 6 trainees is recommended.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.03 – CLASSIFY SURFACE VESSELS

LG 002.02.03 – Classify Surface Vessels		
Location Conditions	**Instructional Sequence**	
SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.02.02	
Time	**Classification**	**Instructor**
100 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Identify specific characteristics of surface vessels to include: a. Acoustic signature characteristics b. Prime mover c. Drive systems d. Blade/shaft configuration e. Role/purpose f. Power source g. Hydrodynamic/transient sources h. SRCs/Swaths i. Frequency characteristics j. Known unknown noise sources 2) Classify surface vessels	1) Individual Problem Solving 2) Lecture 3) EDI 4) Group Problem Solving	
Scenario	Nil.	
Activity Steps	1) Instructor will conduct an icebreaker individual problem solving activity by providing the class with an example LOFAR gram with a contact/target of interest. Instructor will ask trainees to provide their possible answers/solutions to the contact. Instructor will provide feedback to trainees. (20 mins – Method 1) 2) Instructor to conduct lecture to cover LOs. Embedded within the lecture are example sound profiles and LOFAR grams where instructor will explain and demonstrate. (50 min – Method 2 & 3) 3) Instructor will conduct a group problem solving analysis activity. Instructor will provide a LOFAR gram(s) to the class or groups to analyze together. Trainees will have 15 minutes to analyze the LOFAR gram(s) and then present their answers/contact classifications to the class. Instructor will provide feedback. (30 mins – Method 4)	
Instructor Resources	**References**	
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers	1) CFCD 113 Volumes 1 & 2	
Note(s) to Instructor		

LG 002.02.03 – Classify Surface Vessels
1) Examples are to include all available current surface vessels. 2) Historically this has been a challenging skill, anticipate that coaching will be required. 3) Instructor may determine the number of LOFAR grams used in the group problem solving analysis activity. Examples are to include all available current weapons systems. 4) 1 instructor per 6 trainees is recommended.

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.04 – CLASSIFY SURFACE VESSELS LAB

LG 002.02.04 – Classify Surface Vessels Lab		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.03
Time	**Classification**	**Instructor**
250 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Analyze LOFAR grams (surface vessels) 2) Identify surface vessels by passive acoustic data 3) Classify passive acoustic data (surface vessels)		1) Practical (Scenario Based Learning)
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various LOFAR gram examples where they must utilize previous learned skills IOT classify surface vessels. 2) Trainees will be provided 10 contacts (minimum), where there will be a reduction in assistance as they progress in skill. (2 contacts per period for a total of 4 periods of practical, additional 50 min allocated for debrief after entire lab complete). (250 mins – Method 1) 3) After lab is complete, instructor to take up the activity, providing feedback and Directing Staff solution for the contacts conducted. Throughout trainee work period, instructor to walk throughout and provide assistance/feedback.	
Instructor Resources		**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets for trainees 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Examples are to include all available current surface vessels. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.05 – CLASSIFY DIESEL SUBMARINES

LG 002.02.05 – Classify Diesel Submarines		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.04
Time	**Classification**	**Instructor**
100 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Identify specific characteristics of diesel submarines to include: a. Acoustic signature characteristics b. Prime mover c. Drive systems d. Blade/shaft configuration e. Role/purpose f. Power source g. Hydrodynamic/transient sources h. SRCs/Swaths i. Frequency characteristics j. Known unknown noise sources 2) Classify diesel submarines		1) Individual Problem Solving 2) Lecture 3) EDI 4) Group Problem Solving
Scenario	Nil.	
Activity Steps	1) Instructor will conduct an icebreaker individual problem solving activity by providing the class with an example LOFAR gram with a contact/target of interest. Instructor will ask trainees to provide their possible answers/solutions to the contact. Instructor will provide feedback to trainees. (20 mins – Method 1) 2) Instructor to conduct lecture to cover LOs. Embedded within the lecture are example sound profiles and LOFAR grams where instructor will explain and demonstrate. (50 min – Method 2 & 3) 3) Instructor will conduct a group problem solving analysis activity. Instructor will provide a LOFAR gram(s) to the class or groups to analyze together. Trainees will have 15 minutes to analyze the LOFAR gram(s) and then present their answers/contact classifications to the class. Instructor will provide feedback. (30 mins – Method 4)	
Instructor Resources		**References**
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Examples are to include all available current diesel submarines. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.06 – CLASSIFY DIESEL SUBMARINES LAB

LG 002.02.06 – Classify Diesel Submarines Lab		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.05
Time	**Classification**	**Instructor**
250 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Analyze LOFAR grams (diesel submarines) 2) Identify diesel submarines by passive acoustic data 3) Classify passive acoustic data (diesel submarines)		1) Practical (Scenario Based Learning)
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various LOFAR gram examples where they must utilize previous learned skills IOT classify diesel submarines. 2) Trainees will be provided 10 contacts (minimum), where there will be a reduction in assistance as they progress in skill. (2 contacts per period for a total of 4 periods of practical, additional 50 min allocated for debrief after entire lab complete). (250 mins – Method 1) 3) After lab is complete, instructor to take up the activity, providing feedback and DS solution for the contacts conducted. Throughout trainee work period, instructor to walk throughout and provide assistance/feedback.	
Instructor Resources		**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets for trainees 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Examples are to include all available current diesel submarines. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.07 – CLASSIFY NUCLEAR SUBMARINES

LG 002.02.07 – Classify Nuclear Submarines		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.06
Time	**Classification**	**Instructor**
200 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Identify specific characteristics of nuclear submarines to include: a. Acoustic signature characteristics b. Prime mover c. Drive systems d. Blade/shaft configuration e. Role/purpose f. Power source g. Hydrodynamic/transient sources h. SRCs/Swaths i. Frequency characteristics j. Known unknown noise sources 2) Classify nuclear submarines		1) Individual Problem Solving 2) Lecture 3) EDI 4) Group Problem Solving
Scenario	Nil.	
Activity Steps	1) Instructor will conduct an icebreaker individual problem solving activity by providing the class with an example LOFAR gram with a contact/target of interest. Instructor will ask trainees to provide their possible answers/solutions to the contact. Instructor will provide feedback to trainees. (50 mins – Method 1) 2) Instructor to conduct lecture to cover LOs. Embedded within the lecture are example sound profiles and LOFAR grams where instructor will explain and demonstrate. (100 min – Method 2 & 3) 3) Instructor will conduct a group problem solving analysis activity. Instructor will provide a LOFAR gram(s) to the class or groups to analyze together. Trainees will have 15 minutes to analyze the LOFAR gram(s) and then present their answers/contact classifications to the class. Instructor will provide feedback. (50 mins – Method 4)	
Instructor Resources		**References**
1) PowerPoint lecture (embedded examples) 2) SECRET workstation 3) Projector 4) Speakers		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Examples are to include all available current nuclear submarines. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

CO NTDC(A) Approval - 30/11/2021
CNPTG Approval - 08/01/2022
Version No: 3.0

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.08 – CLASSIFY NUCLEAR SUBMARINES LAB

LG 002.02. 08 – Classify Nuclear Submarines Lab		
Location Conditions	**Instructional Sequence**	
SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.02.07	
Time	**Classification**	**Instructor**
500 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Analyze LOFAR grams (nuclear submarines) 2) Identify nuclear submarines by passive acoustic data 3) Classify passive acoustic data (nuclear submarines)	1) Practical (Scenario Based Learning)	
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various LOFAR gram examples where they must utilize previous learned skills IOT classify diesel submarines. 2) Trainees will be provided 10 contacts (minimum), where there will be a reduction in assistance as they progress in skill. (2 contacts per period for a total of 4 periods of practical, additional 50 min allocated for debrief after entire lab complete). (500 mins – Method 1) 3) After lab is complete, instructor to take up the activity, providing feedback and DS solution for the contacts conducted. Throughout trainee work period, instructor to walk throughout and provide assistance/feedback.	
Instructor Resources	**References**	
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets for trainees 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution	1) CFCD 113 Volumes 1 & 2	
Note(s) to Instructor		
1) Examples are to include all available current diesel submarines. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
LG 002.02.09 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS LABS

LG 002.02.09 – Perform Basic Passive Acoustic Analysis Labs		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After LG 002.02.09
Time	**Classification**	**Instructor**
1400 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Analyze passive acoustic data		1) Practical
Scenario	Nil.	
Activity Steps	1) Trainees will be provided various LOFAR gram examples where they must utilize previous learned skills IOT classify contacts. 2) Trainees will be provided 4 mixed contact labs consisting of a minimum of 80 contacts per lab (minimum one weapons system, one surface vessel, one diesel submarine and one nuclear submarine per lab). Throughout trainee work period, instructor to walk throughout and provide assistance/feedback. (300 mins per lab) (1200 mins – Method 1) 3) After each lab is completed, instructor to take up the activity, providing feedback and Directing Staff solution for the contacts conducted, (50 mins per lab). Instructor may also review and have trainees practice any lab contacts that are deemed areas of weakness and/or concern. (200 mins total) 4)	
Instructor Resources		**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Headsets 1 per trainee 5) Chronoscope/stopwatch 6) Calculator 7) Acoustic files		1) CFCD 113 Volumes 1 & 2
Note(s) to Instructor		
1) Examples are to include all available current submarines. 2) Historically this has been a challenging skill, anticipate that coaching will be required.		

EO 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS
EC 002.02 – PERFORM BASIC PASSIVE ACOUSTIC ANALYSIS

EC 002.02 – Perform Basic Passive Acoustic Analysis		
Scope	**Location Conditions**	**Instructional Sequence**
PO 002	SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.02.09
Time	**Classification**	**Instructor**
250 min	SECRET FVEY	RQ-MS SONAR OP
Scenario	Nil.	
Assessment Guidance	**Input:** Trainees will be provided 6 contacts to analyze (minimum one weapons system, one surface vessel, one diesel submarine and one nuclear submarine) and all resources listed. **Performance:** Trainees will conduct acoustic analysis IOT classify 6 contacts. (200 mins) **Output:** IAW ADAC passive acoustic analysis checklist, trainees must obtain 75% for SAT on EC 002.02 Classification of contact must be 80% for SAT on EC 002.02 **Feedback:** Provided immediate verbal and written feedback IAW EC 002.02 checklist upon completion of the EC. (50 mins for debrief)	

Resources	**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets/trainee 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution	1) CFCD 113 Volumes 1 & 2

Note(s) to Instructor	
1) All notes material is classified material and must be stored and destroyed IAW NDSODs.	
Assessment Limitations	No assistance provided (unlike on ship where you may have support).
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

UNCLASSIFIED

Contact Number: _____ **Contact Location:** _____ **Measurement Date / Time:** _____

Sensor: _____ **Sensor Depth (ft):** _____ **SLD (ft):** _____ **AN:** _____

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPPM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry... Ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2% tolerance	
Class Ind/ID Swath	
1	1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Shaft and Engine Related Sources

Drive Mode	RR
D/E	

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed
Class Indicators		
1	1	1
Modulation/Banding		
1	1	1
Others		
	1	

Transients, Significant Events, Tactical Info

Time	Details
	1 pt for each *significant* event listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1
1/2 harmonics must be listed		
Augmentors	1	
for the frequency and calling it an augmentor		
Auxiliaries		
1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter		
Others	1	
1pt awarded to other significant sources		
Unknowns	0	

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1
SRCRuns/Trails	

Aural Characteristics

Time	Details
	1 pt for each significant aural characteristic listed

Classification

4 points total

1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

PROTECTED B (when complete)

PO 002 – ANALYZE PASSIVE ACOUSTIC DATA
LG 002.02.10 – PERFORM TEAM BASIC PASSIVE ACOUSTIC ANALYSIS

LG 002.02.10 – Perform Basic Passive Acoustic Analysis		
Location Conditions		**Instructional Sequence**
SECRET Classroom (ADAC classroom/mobile classroom)		After EC 002.02
Time	**Classification**	**Instructor**
350 min	SECRET FVEY	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Conduct team passive acoustic data analysis 2) Provide command with tactical assessment		1) Scenario Based Learning
Scenario	ADAC will provide tactical scenario.	
Activity Steps	1) Instructor will break the class into groups (2-4 trainees per group). 2) Trainees will be provided a tactical scenario (1 scenario per group) utilizing the passive ASW equipment logs IAW NCPM 600. 3) Trainees will be provided a lab with minimum of 4 contacts. All contacts will be configured and displayed to emulate CANTASS appearance and frequency spectrums. Lab will include: a. 1 x Allied contact b. 2 x Submarine threat contacts c. 1 x Weapon 4) Instructor will first provide a demonstration of the command classification tool IAW ATP 28 Vol 2 and/or AXP-05 EXTAC 198 as they apply to the scenario and classify the first initial contact to ensure group understanding and expectations for next activity. (50 mins) 5) Groups will then analyze the contacts in the lab and using the command classification tool provided, identify the criteria to achieve the command classification of POSSUB High 4 IAW ATP-28 and/or AXP-05 EXTAC 198 for each subsurface threat target. Throughout trainee work period, instructor to walk throughout and provide coaching with decreased assistance/feedback as labs progress. (250 mins – Method 1) 6) Upon completion of the group analysis, each group will present their answers to the instructor(s). Instructor(s) will provide each group with feedback. (50 mins)	
Instructor Resources		**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Headsets 1 per trainee 5) Chronoscope/stopwatch 6) Calculator 7) Acoustic files 8) Tactical Scenario (1 per group, maximum of 4 scenarios) 9) Command Classification Tool		1) CFCD 113 Volumes 1 & 2 2) ATP 28 Vol 2 3) AXP-05 EXTAC 198

LG 002.02.10 – Perform Basic Passive Acoustic Analysis	
10) NCPM 600 Logs 11) Instructor DS solution	
Note(s) to Instructor	
1) Scenario based activity be conducted upon completion of EC 002.02 but prior to PC 002. 2) Instructor to ensure maximum participation of all trainees and guide focus on analysis and justification.	

PO 002 – ANALYZE PASSIVE ACOUSTIC DATA
PC 002 – ANALYZE PASSIVE ACOUSTIC DATA

PC 002 – Analyze Passive Acoustic Data		
Scope	**Location Conditions**	**Instructional Sequence**
PO 002	SECRET Classroom (ADAC classroom/mobile classroom)	After LG 002.02.10
Time	**Classification**	**Instructor**
250 min	SECRET FVEY	RQ-MS SONAR OP

Scenario	N/A
Assessment Guidance	**Input:** Trainees will be provided 6 contacts to analyze (minimum one weapons system, one surface vessel, one diesel submarine and one nuclear submarine) and all resources listed. **Performance:** Trainees will conduct acoustic analysis IOT classify 6 contacts. (200 mins) **Output:** IAW ADAC passive acoustic analysis checklist, trainees must obtain 75% for SAT on PC 002. Classification of contact must be 80% for SAT on PC 002. **Feedback:** Provided immediate verbally and written IAW PC 002 checklist upon completion of the PC. (50 mins for debrief)

Resources	**References**
1) Individual SECRET workstation/trainee 2) Projector 3) Speakers (instructor) 4) Acoustic worksheets 5) Acoustic files 6) Headsets/trainee 7) Chronoscope/stopwatch 8) Calculator 9) Instructor DS solution	1) CFCD 113 Volumes 1 & 2

Note(s) to Instructor	
1) All note material is classified material and must be stored and destroyed IAW NDSODs.	
Assessment Limitations	No assistance provided (unlike on ship where you may have support).
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

UNCLASSIFIED

Contact Number: _____ **Contact Location:** _____ **Measurement Date / Time:** _____

Sensor: _____ **Sensor Depth (ft):** _____ **SLD (ft):** _____ **AN:** _____

Tactical Data

CPA Time	00:01	1
Contact Aspect		1
Contact Depth (ft)	50%	1
CPA Range (yds)	50%	1
Doppler Speed (kts)	2	1
TPK Speed (kts)	1	1
Turns Per Knot	1	1
Bow ADR (yds)		
Bow ADR Freq		
Stern ADR (yds)		
Stern ADR Freq		
Best Tracking		
Best Detection		

Propulsion

Blade Rate	# Blades	Shaft Rate	SPPM	Turbine	Pred. Harmonics	CFR	CSR	EFR	ERPM	Pred. Harm.
1pt for each different entry... ie. If the contact has two shaft rates th										
0.5		0.2	12	0.2	Shaft, Blade, Turbine	0.1	0.2	0.5	12	1/2 to be listed
1	1	1	1	1	1pt for each type listed	1	1	1	1	1

Swaths

Name	Fund.
2% tolerance	
Class Ind/ID Swath	
1	1

1 pt for the first Swath listed and 1 pt for every 3rd thereafter

Shaft and Engine Related Sources

Drive Mode	RR
D/E	

Source	Fund.	Pred. Harm.
	0.5	1/2 to be listed
Class Indicators		
1	1	1
Modulation/Banding		
1	1	1
Others		
	1	

Transients, Significant Events, Tactical Info

Time	Details
	1 pt for each *significant* event listed

Auxiliaries, Augmentors & Unknowns

Source	Fund.	Harm.
	0.2	
Class Indicators		
1	1	1

1/2 harmonics must be listed

Augmentors 1
for the frequency and calling it an augmentor

Auxiliaries
1 pt for the first auxiliary listed and 1 pt for every 3rd thereafter

Others 1
1pt awarded to other significant sources

Unknowns 0

Speed Related Components

Name	Fundamental
	0.5 (tolerance)
1	1

Aural Characteristics

Time	Details
	1 pt for each significant aural characteristic listed

Classification

4 points total

1pt for surf/subsurf, nuclear/diesel, type, class

Justification

1 pt iff the information justifies the classification

Analysis Summary

Secondary Contact

This Section is used by instructors to indicate directions to the Trainees for the completion of theGram to be Analyzed.

ANALYSIS	CLASSIFICATION

PROTECTED B (when complete)

PO 003 – OPERATE COMMAND SYSTEM

1. **Performance**: Operate Command System (B0050).

2. **Conditions**:

   a. Given:

      (1) Command System Display,
      (2) Blank SPS Logs,
      (3) Scientific Calculator,
      (4) Scenario,
      (5) Sonobuoy Processing System, and
      (6) References,

   b. Denied: Nil; and

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at sea.

3. **Standard**: The acoustic operator shall Operate Command System (B0050) by:

   a. Recording acoustic intelligence data (F0010) by:

      (1) Maintaining stateboards (B0015),
      (2) Annotating recording media (C0075), and
      (3) Maintaining operational logs (B0055), and

   b. Operating passive acoustic equipment (C0020) by:

      (1) Operating Sonobuoys Processing System (SPS) (C0215),
      (2) Monitoring sonobuoys (C0060),
      (3) Tracking acoustic contacts (C0070),
      (4) Localizing passive acoustic contacts (C0085), and
      (5) Amplifying track information (E0020).

4. **Training Limitations**: Nil.

EO 003.01 – RECORD ACOUSTIC INTELLIGENCE DATA
LG 003.01.01 – OPERATE SPS RECORDER

LG 003.01.01 – Operate SPS Recorder		
Location Conditions	**Instructional Sequence**	
HMC Ship or ADAC Classroom		
Time	**Classification**	**Instructor**
500 min	SECRET	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Login and setup recorder 2) Operate recorder 3) Extract files	1) EDI 2) Practical	
Scenario		
Activity Steps	1) Instructor will demonstrate the login and setup procedures for the SPS data recorder to all trainees at once on projector. Trainees will then be asked to recreate those steps individually to confirm they understand the process on their own. (200 min – Method 1) 2) Instructor will demonstrate how to operate the recorder. (25 min – Method 1) 3) Trainees will then operate the recorder IAW Acoustic Data Recorder (ADR) checklist. (70 min – Method 2) 4) Instructor will demonstrate how to extract files. (25 min – Method 1) 5) Trainee will practice the process of extracting files. (180 min – Method 2)	
Instructor Resources	**References**	
1) SPS recorder 2) Acoustic Data Recorder (ADR) checklist 3) Hard-drive dock 4) Stand-alone Secret computer or laptop	1) Acoustic Data Recorder Operation and Maintenance Manual (REACH)	
Note(s) to Instructor		
Nil.		

EO 003.01 – RECORD ACOUSTIC INTELLIGENCE DATA
EC 003.01 – RECORD ACOUSTIC INTELLIGENCE DATA

EC 003.01 – Record Acoustic Intelligence Data		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.01	HMC Ship or ADAC Classroom	After LG 003.01.01
Time	**Classification**	**Instructor**
300 min (25 mins per trainee)	SECRET	RQ-MS SONAR OP

Scenario	Nil
Assessment Guidance	**Input:** Trainee will be onboard a HMC Ship or in an ADAC classroom, and provided an ADR, hard-drive dock and a stand-alone secret computer or laptop. **Performance:** Trainee must operate ADR and demonstrate ability to setup, operate and extract files IAW EC 003.01 checklist. (20 mins per trainee) **Output:** Trainee must achieve 8/ 10 SATs pass this EC. **Feedback:** Immediate verbal and written feedback IAW EC 003.01 checklist. (5 mins per trainee)

Instructor Resources	**References**
1) SPS ADR recorder 2) ADR Checklist 3) Hard-drive dock Stand-alone Secret computer or laptop	1) Acoustic Data Recorder Operation and Maintenance Manual (REACH)

Note(s) to Instructor
1) All notes material is classified material and must be stored and destroyed IAW NDSODs.

Assessment Limitations	Availability of HMC Ship or ADAC classroom.
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

EC 003.01 RECORD ACOUSTIC INTELLIGENCE DATA			
Item	**SAT**	**UNSAT**	**Remarks**
Flash up ADR	☐	☐	
Login to ADR	☐	☐	
Insert and mount ADR hard drive	☐	☐	
Setup for playback	☐	☐	
Setup for record to include:			
Select Rewind hard drive	☐	☐	
Press Record button	☐	☐	
Press Stop Record	☐	☐	
Unmount ADR hard drive	☐	☐	
Log out	☐	☐	
Shut down ADR	☐	☐	

Score: /10

The trainee must achieve 8/10 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 003.02 – OPERATE PASSIVE ACOUSTIC EQUIPMENT
LG 003.02.01 – OPERATE SPS

LG 003.02.01 – Operate SPS		
Location Conditions		**Instructional Sequence**
SEACOT		After EC 003.01
Time	**Classification**	**Instructor**
1500 min	SECRET	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Conduct initial contact procedure 2) Update/amplify your track information 3) Conduct lost contact procedure 4) Conduct regain contact procedure 5) Annotate logs 6) Update/maintain CMS 330 stateboards		1) EDI 2) Practical
Scenario		
Activity Steps	1) Instructor will explain and demonstrate how to conduct initial contact procedures to include voice procedures. (15 min – Method 1) 2) Trainees will then pair-up and practice initial contact including voice procedures together. (35 min – Method 2) 3) Instructor will explain and demonstrate how to update/amplify contact information procedures to include voice procedures. (15 min – Method 1) 4) Trainees will then pair-up and practice updating/amplifying contact information including voice procedures together. (35 min – Method 2) 5) Instructor will demonstrate the lost contact procedure including voice procedures. (15 min – Method 1) 6) Trainees will then pair-up and practice lost contact including voice procedures together. (35 min – Method 2) 7) Instructor will demonstrate the regain contact procedure including voice procedures. (15 min – Method 1) 8) Trainees will then pair-up and practice regain contact including voice procedures together. (35 min – Method 2) 9) Instructor will demonstrate how to fill out a SPS log and provide an example of a completed log. (25 min – Method 1) 10) Trainees will annotate a log with information provided by handout. (25 min – Method 2) 11) Trainees will be shown how to update the CMS 330 stateboards and apply changes (50 min – Method 1)	
Instructor Resources		**References**
1) Voice procedures handout 2) Go-operator 3) Log handout 4) DS log handout solution 5) SPS job sheet		1) NCPM 600 2) Sonobuoy Reference Guide

LG 003.02.01 – Operate SPS
Note(s) to Instructor
Nil.

EO 003.02 – OPERATE PASSIVE ACOUSTIC EQUIPMENT
EC 003.02.01 – OPERATE PASSIVE ACOUSTIC EQUIPMENT

EC 003.02 – Operate Passive Acoustic Equipment		
Scope	**Location Conditions**	**Instructional Sequence**
LG 003.02.01	SEACOT	After LG 003.02.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-MS SONAR OP

Scenario	N/A
Assessment Guidance	**Input:** Trainee will be in a SEACOT and provided a scenario with the following game pieces: • 1 Subsurface contact • Minimum of 3 Sonobuoys to monitor **Performance:** Trainee must operate SPS, monitor sonobuoys, detect, track and provide contact reports to SCS / instructor demonstrating the ability to Operate SPS IAW EC 003.02.01 checklist. (20 mins per trainee) **Output:** Trainee must achieve 20 / 25 SATs on to pass this assessment. **Feedback:** Immediate verbal and written feedback IAW EC 003.02.01 checklist. (5 mins per trainee)

Instructor Resources	References
1) EC scenario 2) Go-operator 3) SCS / Instructor 4) SPS Job Sheet 5) SPS Log template	1) NCPM 600 2) Sonobuoy Reference Guide 3) CFCD 113 Vol 1 & 2

Note(s) to Instructor
Ensure trainee conducts a Watch Turnover to either instructor / trainee.

Assessment Limitations	SEACOT is unable to recreate the contact fidelity and oceanographic environment that would be observed onboard in real time operation. Some aspects will require simulation.
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

EC 003.02.01 OPERATE PASSIVE ACOUSTIC EQUIPMENT (SPS)			
Item	**SAT**	**UNSAT**	**Remarks**
Login under SPSO role.	☐	☐	
Request control on main display	☐	☐	
Conduct initial contact report to SPS to include:			
Create acoustic bearing on strongest frequency on either RF channel	☐	☐	
Conduct initial contact report to SPS to include; • RF channel # • Frequency holding	☐	☐	
Conduct amplification report to SCS to include:			
Up / Down Doppler or steady	☐	☐	
Assessment of contact	☐	☐	
Update contact info in CTC detailed window to include:			
Surface/sub-surface (if applicable)	☐	☐	
Friendly/threat (if applicable)	☐	☐	
Country (if applicable)	☐	☐	
Conduct lost contact report to SCS to include:			
Last known bearing on either RF channel	☐	☐	
CTC fade or abrupt out	☐	☐	
Assessment of contact	☐	☐	
Assign a contact lost into the CTC management update window to the appropriate CTC	☐	☐	
Complete SPS logs to include:			
RF channel:			
Number #	☐	☐	
Buoy type	☐	☐	
Depth	☐	☐	
Life	☐	☐	
CTC details:			
Gained / Lost	☐	☐	
Frequency	☐	☐	
Source (if known)	☐	☐	
Bearing	☐	☐	
Force Track number#	☐	☐	
Remarks:			
CTC / Ownship events	☐	☐	

EC 003.02.01 OPERATE PASSIVE ACOUSTIC EQUIPMENT (SPS)			
System faults	☐	☐	
Conduct Watch Turnover:	☐	☐	

Score: / 25

The trainee must achieve 20 / 25 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 003.02 – OPERATE PASSIVE ACOUSTIC EQUIPMENT
LG 003.02.02 – OPERATE SPS FOR TORPEDO TRACKING PROCEDURES

LG 003.02.02 – Operate SPS for Torpedo Tracking Procedures		
Location Conditions		**Instructional Sequence**
SEACOT		After EC 003.02.01
Time	**Classification**	**Instructor**
1000 min	SECRET	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Conduct initial torpedo report 2) Create torpedo bearing line 3) Monitor and update torpedo bearing line		1) EDI 2) Practical
Scenario		
Activity Steps	1) Instructor will demonstrate the initial Torpedo contact procedures including the voice reports and how to create a torpedo bearing line. Instructor will demonstrate how to monitor and update the bearing of the torpedo. (500 min – Method 1) 2) Trainees will then pair-up and practice their initial contact procedures to include the voice reports to SCS, they will monitor and update the bearing of the torpedo as required. Instructor/Go-operator will alter game piece to ensure there is change in bearing in game scenario. 3) (500 min – Method 2)	
Instructor Resources		**References**
1) Voice Procedures sheet 2) Go-operator		1) NCPM 600 2) CFCD 106 Maritime Tactical Instructions MARTI 2122 3) Sonobuoy Reference Guide
Note(s) to Instructor		
Nil.		

EO 003.02 – OPERATE PASSIVE ACOUSTIC EQUIPMENT
EC 003.02.02 – OPERATE SPS FOR TORPEDO TRACKING PROCEDURES

EC 003.02.02 – Operate SPS For Torpedo Tracking Procedures		
Scope	**Location Conditions**	**Instructional Sequence**
LG 003.02.02	SEACOT	After LG 003.02.02
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-MS SONAR OP

Scenario	
Assessment Guidance	**Input:** Trainee will be in a SEACOT and provided a scenario with the following game pieces: • 1 Subsurface contact • Minimum of 3 Sonobuoys to monitor • Minimum of 1 threat torpedo **Performance:** Trainee must operate SPS, monitor sonobuoys, detect, track and provide torpedo reports to SCS / instructor demonstrating the ability to Operate SPS for torpedo tracking procedures IAW EC 003.02.02 checklist. (20 mins per trainee) **Output:** Trainee must achieve 8 / 10 SATs on non-critical factors **AND** all critical factors to pass this EC. **Feedback:** Immediate verbal and written feedback IAW EC 003.02.02 checklist. (5 mins per trainee)

Instructor Resources	**References**
1) EC scenario 2) Go-operator 3) SCS / Instructor 4) SPS Job Sheet	1) NCPM 600 2) Sonobuoy Reference Guide 3) CFCD 106 Maritime Tactical Instructions 4) CFCD 113 Vole 1 & 2

Note(s) to Instructor	
Nil.	
Assessment Limitations	SEACOT is unable to recreate the contact fidelity and oceanographic environment that would be observed onboard in real time operation. Some aspects will require simulation.
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

EC 003.02.02 OPERATE PASSIVE ACOUSTIC EQUIPMENT (SPS)			
Item	**SAT**	**UNSAT**	**Remarks**
Login under SPSO role.	☐	☐	
Request control on main display	☐	☐	
Conduct Torpedo Tracking Procedures			
Identify Torpedo signature on sonobuoy(s)	☐	☐	***Critical**
Raise the alarm	☐	☐	***Critical**
Create torpedo bearing line on RF channel	☐	☐	
Conduct initial torpedo report to SCS to include:			
Torpedo bearing(s)	☐	☐	
RF channel #	☐	☐	
Monitor and update torpedo bearings with assessments to include:			
Up / Down Doppler / steady	☐	☐	
Fade in / out or Abrupt in/out	☐	☐	
CPA	☐	☐	

Score: / 10

The trainee must achieve 8 / 10 SATs on non-critical factors AND all critical factors to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 003 – OPERATE COMMAND SYSTEM
LG 003.02.03 – OPERATE SPS FOR ASW MISSION

LG 003.02.03 – Operate SPS for ASW Mission		
Location Conditions		**Instructional Sequence**
SEACOT		After EC 003.02.02
Time	**Classification**	**Instructor**
1800 min	SECRET	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Perform duties of SPS during operations		1) Practical
Scenario	Operational mission IAW NWTP Vol 4.	
Activity Steps	1) Trainees will conduct 3 x 50 mins scenario runs (including 10 min debrief) where they will be required to set up the Command System, operate equipment, detect and track sub-surface contact and torpedoes and provide supervisor with reports. (1800 min – Method 1)	
Instructor Resources		**References**
1) Mission scenario 2) Go-operator 3) SPS Job Sheet 4) SCS 5) UWWD/Officer of the Watch		1) NCPM 600 2) CFCD 106 Maritime Tactical Instructions 3) CFCD 113 Vol 1 & 2 4) Sonobuoy Reference Guide
Note(s) to Instructor		
1) Support manning IAW NWTP Vol.2.		

PO 003 – OPERATE COMMAND SYSTEM (SPS)
PC 003 – OPERATE COMMAND SYSTEM (SPS)

PC 003 – Operate Command System		
Scope	**Location Conditions**	**Instructional Sequence**
PO 003	SEACOT	After LG 003.02.03
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-MS SONAR OP

Scenario	Operational mission IAW NWTP Vol 4.
Assessment Guidance	**Input** – Trainee will be in a SEACOT provided a SPS MFW, a subsurface game piece with torpedoes. **Performance** – Trainee must operate/update sub-surface contact and torpedo data and demonstrate the ability to operate SPS during operational mission. (20 mins per trainee) Each run will include: 1) 3 sonobuoys to monitor 2) 1 surface contact 3) 1 submarine 4) 2 torpedoes to track **Output** – Trainee must achieve 24 / 32 SATs on non-critical factors **AND** all critical factors to pass this assessment. **Feedback** – Immediate verbal and written feedback IAW PC 003 checklist. (5 mins per trainee)

Resources	**References**
1) Mission scenario 2) Go-operator 3) SPS Job Sheet 4) SCS 5) UWWD/Officer of the Watch	1) NCPM 600 2) CFCD 106 Maritime Tactical Instructions 3) Sonobuoy Reference Guide

Note(s) to Instructor	
Support manning IAW NWTP Vol.2.	
Assessment Limitations	SEACOT will not provide the acoustic fidelity that SPS is capable of receiving at sea. Some acoustic information will need to be simulated. SEACOT does not have an Acoustic Data Recorder or simulator. Will not be tested in PC 003.
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

PC 003 OPERATE COMMAND SYSTEM (SPS)			
Item	**SAT**	**UNSAT**	**Remarks**
Login under SPSO role.	☐	☐	
Request control on main display	☐	☐	
Conduct initial contact report to SPS to include:			
Create acoustic bearing on strongest frequency on either RF channel	☐	☐	
Conduct initial contact report to SPS to include; • RF channel # • Frequency hold	☐	☐	
Conduct amplification report to SCS to include:			
Up / Down Doppler or steady	☐	☐	
Assessment of contact	☐	☐	
Update contact info in CTC detailed window to include:			
Surface/sub-surface (if applicable)	☐	☐	
Friendly/threat (if applicable)	☐	☐	
Country (if applicable)	☐	☐	
Conduct lost contact report to SCS to include:			
Last known bearing on either RF channel	☐	☐	
CTC fade or abrupt out	☐	☐	
Assessment of contact	☐	☐	
Assign a contact lost into the CTC management update window to the appropriate CTC	☐	☐	
Complete SPS logs to include:			
RF channel:			
Number #	☐	☐	
Buoy type	☐	☐	
Depth	☐	☐	
Life	☐	☐	
CTC details:			
Gained / Lost	☐	☐	
Frequency	☐	☐	
Source	☐	☐	
Bearing	☐	☐	
Force Track number #	☐	☐	
Remarks:			
CTC / Ownship events	☐	☐	

PC 003 OPERATE COMMAND SYSTEM (SPS)			
System faults	☐	☐	
Conduct Torpedo Tracking Procedures			
Identify Torpedo signature on sonobuoy(s)	☐	☐	***Critical**
Raise the alarm	☐	☐	***Critical**
Create torpedo bearing line on RF channel	☐	☐	
Conduct initial torpedo report to SCS to include:			
Torpedo bearing	☐	☐	
RF channel #	☐	☐	
Monitor and update torpedo bearings with assessments to include:			
Up / Down Doppler / steady	☐	☐	
Fade in / out or Abrupt in/out	☐	☐	
CPA	☐	☐	

Score: / 32

The trainee must achieve 24 / 32 SATs on non-critical factors AND all critical factors to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 004 – CONDUCT TARGET MOTION ANALYSIS (TMA)

1. **Performance**: Conduct Target Motion Analysis (TMA) (C0255).

2. **Conditions**:

   a. Given:

      (1) Parallel or rolling ruler,
      (2) Scientific Calculator,
      (3) Manoeuvring boards,
      (4) 10-point divider,
      (5) References,
      (6) TMA aide memoire
      (7) Scenarios, and
      (8) Coloured pencils, and
      (9) Supervision.

   b. Denied: Nil; and

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at sea.

3. **Standard**: The acoustic operator shall conduct TMA (C0255) by:

   a. Operating CMS for TMA by:

      (1) Setting up TMA plot (C0086),
      (2) Plotting TMA data (C0090),
      (3) Performing TMA calculations (C0087) in system, and
      (4) Constructing contact AOP (C0100), and

   b. Validating TMA data by:

      (1) Analyzing TMA data (C0095), and
      (2) Identifying changes in contact readiness and tactical states (F0015).

4. **Training Limitations**: Nil.

EO 004.01 – OPERATE CMS FOR TMA
LG 004.01.01 – OPERATE CMS FOR TMA

LG 004.01.01 – Operate CMS for TMA		
Location Conditions		**Instructional Sequence**
SEACOT (preferred) SEAWOLF (secondary)		After EC 004.02
Time	**Classification**	**Instructor**
600 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Setting up TMA TDA in CMS 2) Initiate system TMA plot 3) Recognize strip plot relationships 4) Publish AOP		1) EDI 2) Practical
Scenario		
Activity Steps	1) Instructor to explain and demonstrate the sequence of events to set up, initiate TMA and recognizing the strip plot relationship before publishing the AOP. (30 min – Method 1) 2) Trainees will then be put at their own consoles where they will be asked to conduct the LOs on their own with instructor support (instructor use projector of their screen and walk trainees through). (70 min – Method 1) 3) Trainees are then asked to conduct the LOs individually using the training aid checklist to walk through the required steps. Contacts will be input to the system where trainees will be required to use CMS for TMA. Each trainee must be provided at least 3 scenarios for training. Feedback to be provided throughout. (100 min – Method 2)	
Instructor Resources		**References**
1) SEACOT MFW in SPS role 2) SEACOT CANTASS console 3) TMA checklist 4) Instructor projector for console 5) Paper / pencil 6) Scientific Calculator, 7) Maneuvering boards		1) CMS 330 Combat System User Manual (CSUM)
Note(s) to Instructor		
1) Scenarios will be one contact per run, and in each scenario either own ship or contact must alter course.		

EO 004.01 – OPERATE CMS FOR TMA
EC 004.01 – OPERATE CMS FOR TMA

EC 004.01 – Operate CMS for TMA		
Scope	**Location Conditions**	**Instructional Sequence**
PO 004	SEACOT (preferred) SEAWOLF (secondary)	After LG 004.01.01
Time	**Classification**	**Instructor**
300 min	UNCLASS	RQ-MS SONAR OP

Scenario	N/A
Assessment Guidance	**Input:** Trainee will be at a CMS console in SPS or SCS role, provided a subsurface contact supplied through SEACOT CANTASS. Trainee may use manual plotting / calculation techniques (paper/pencil, calculator, manoeuvring boards) during the assessment. **Performance:** Trainee must operate CMS to run TMA solution, verbally indicate the contact relationship in a geo strip plot, and create an AOP IAW EC 004.01 checklist. Manual TMA solutions may be used to support solutions. (20 min per trainee) **Output:** Trainee must achieve 17 / 23 SATs on non-critical factors to pass this EC. **Feedback:** immediate verbal and written feedback IAW EC 004.01 checklist. (5 min per trainee)

Resources	**References**
1) SEACOT MFW in SPS role 2) SEACOT CANTASS console 3) TMA checklist 4) Instructor projector for console 5) Paper / pencil 6) Scientific Calculator, 7) Maneuvering boards	1) CMS 330 Combat System User Manual (CSUM)

Note(s) to Instructor	
1) Two trainees can be assessed concurrently by one instructor.	
Assessment Limitations	Trainer buttonology differs from ship layout.
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

EC 004.01 OPERATE CMS FOR TMA			
Item	**SAT**	**UNSAT**	**Remarks**
Login under SPSO role	☐	☐	
Create and calculate a TMA TDA solution to include:			
Select Tactical Operations tab	☐	☐	
Select Tactical Decision Aids	☐	☐	
TDA manager select the UWW button	☐	☐	
Select Target Motion Analysis	☐	☐	
Hook Bearing Line and Select TACPLOT entry	☐	☐	
Select Global TMA configuration	☐	☐	
Input given ranges then Save and Exit	☐	☐	
Select Ignore CANTASS Track History	☐	☐	
Select Publish AOP and STM	☐	☐	
Select save and compute	☐	☐	
Publish TMA data to include:			
Confirm TDA is displayed in the TAC right column with the title CCS tracking trial	☐	☐	
Operator record the TMA ID # (i.e., 4993493066)	☐	☐	
Select Picture Compilation/TMA line	☐	☐	
Verify the TMA ID number matches the one recorded above	☐	☐	
Verify CANTASS track and its ambiguous bearing lines are being plotted	☐	☐	
Select the solutions tab to observe the graph being plotted	☐	☐	
Publish AOP to STM box is checked	☐	☐	
Resolve for ambiguity procedures to include:			
After manoeuver verify false bearing eliminated from solution	☐	☐	
Select track with bearing ambiguity (i.e., 1001-2001);	☐	☐	
Select BRG1 or BRG2 (true bearing xxx) at the CANTASS adapter presentation panel	☐	☐	
Select as “Resolve as true bearing” button	☐	☐	
Record time	☐	☐	

EC 004.01 OPERATE CMS FOR TMA		
Score: / 23 **The trainee must achieve 17 / 23 SATs to pass this assessment.**		
Positive Feedback	**Improvement Feedback**	
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 004.02 – VALIDATE TMA DATA
LG 004.02.01 – LINE OF SOUND

LG 004.02.01 – Line of Sound		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		Prior to LG 004.01.01
Time	**Classification**	**Instructor**
500 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Understand line of sound diagram 2) Complete calculations for LOS 3) Create line of sound diagram		1) Lecture 2) EDI 3) Scenario Based Learning
Scenario	Nil.	
Activity Steps	1) Instructor to explain a completed line of sound diagram on the white board, including all acronyms and calculations. Instructor must identify how to determine the missing information, and how to conduct the calculations with the information provided. Trainees to be provided a handout that specifies the acronyms and calculations. (100 minutes) 2) Instructor to hand out scenario sheet for trainees to conduct in pairs/threes to practice calculations. First sheet to include 4 questions. (50 min) 3) Trainees are then provided handouts where they will individually practice calculations. Minimum of 8 problems to be covered, increasing in complexity throughout the lesson. Instructor to provide assistance throughout, and upon completion of each activity sheet, take up with the class. (150 minutes) 4) Instructor to conduct a lesson on how to utilize the LOS calculations and displaying it on a maneuvering board. (20 min) 5) Partial information given to trainees to complete line of sound diagram using calculations, results to be displayed on maneuvering board. (180 minutes)	
Instructor Resources		**References**
1) White board & markers 2) LOS example (for lesson) 3) LOS worksheets (12 questions total) 4) Maneuvering boards 5) Calculators/per trainee 6) Scrap paper (calculations) 7) Pencils 8) Rolling / parallel rulers		1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook
Note(s) to Instructor		
1) Review LOS example prior to instruction.		

EO 004.02 – VALIDATE TMA DATA
LG 004.02.02 – DOPPLER THEORY & FREQUENCY RELATIONSHIPS

LG 004.02.01 – Line of Sound		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		After LG 004.02.01
Time	**Classification**	**Instructor**
150 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Application of Doppler theory 2) Understand difference in frequency relationships a. Frequency observed b. Frequency received c. Frequency corrected		1) Think/Pair/Share 2) Lecture 3) Video 4) Table Top
Scenario	Nil.	
Activity Steps	1) Instructor to ask each trainee to come up with their own explanation of Doppler theory. Trainees are given 4 min to create their own definition. (4 min) 2) Trainees are then asked to pair up with another trainee, share their definition and come up with one as a pair. (4 min) 3) Pairs are to present their definition to the class (4 min) 4) Instructor to compile the salient points of each of the definitions provided and lead a discussion on the important factors that contribute to TMA regarding Doppler theory. (8 min) 5) Instructor to conduct a lesson on key concepts of Doppler theory to include a video. (30 minutes) 6) Instructor to carry on with lecture into the frequency relationships, providing examples of Doppler theory and its effect on acoustic frequencies. (15 min) 7) Instructor to lead a table top using the magnetic fleet in a box, depicting the different relationships using white board markers to represent aspects and situations (opening and closing). (35 min) 8) Instructor to demonstrate how to interpret a time frequency plot and relationships. (20 min) 9) Trainees are then broken into pairs and given worksheets where they must interpret time and frequency/relationship problems, (25 min) 10) Instructor to provide feedback on the worksheets and answer any remaining questions (5 min)	
Instructor Resources		**References**
1) Magnetic fleet in a box 2) White board and markers 3) Doppler theory/relationships Presentation (including video) 4) Time frequency worksheets 5) Time frequency example (instructor)		1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook

LG 004.02.01 – Line of Sound
Note(s) to Instructor
1) Historically this has been a difficult concept for trainees.

EO 004.02 – VALIDATE TMA DATA
LG 004.02.03 – RANGE CALCULATIONS & RELATIVE VELOCITY

LG 004.02.01 – Line of Sound		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		After LG 004.02.02
Time	**Classification**	**Instructor**
100 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Understand requirements of range calculations for: a. Single leg Ekelund b. Double leg Ekelund c. Frequency rate Ekelund d. Warren ranging 2) Understand RelVel problems a. True course b. CPA		1) Lecture 2) Demonstration 3) Group discussion
Scenario	Nil.	
Activity Steps	1) Instructor to conduct presentation/lesson on range calculations. Trainees must understand the application of the calculations and how it applies in verifying information in CMS. Trainees may have opportunity to complete some questions. (70 min) 2) Instructor to conduct lesson on the introductive theory to RelVel. Utilize videos to support the theory prior to instruction on how to conduct RelVel problem solving. (25 min) 3) Instructor to conduct wrap up activity, where trainees are encouraged to discuss how the information of the lesson applies in the performance of their job. (5 min)	
Instructor Resources		**References**
1) PowerPoint presentation 2) DWAN or standalone computer 3) Projector 4) Maneuvering boards 5) Calculators/per trainee 6) Scrap paper (calculations) 7) Pencils 8) Rolling / parallel rulers		1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook
Note(s) to Instructor		
Nil.		

EO 004.02 – VALIDATE TMA DATA
LG 004.02.04 – GEOGRAPHIC STRIP PLOT

LG 004.02.04 – Geographic Strip Plot		
Location Conditions		**Instructional Sequence**
Unclassified Classroom		After LG 004.02.03
Time	**Classification**	**Instructor**
50 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Recognize geographic strip plots a. Bearing rate b. Fan pattern c. Intersecting		1) EDI 2) Collaborative problem solving
Scenario	Nil.	
Activity Steps	1) Instructor to conduct a lesson on geographic strip plot in order for trainees to identify directionality of travel and relationship. Examples are to be provided and discussed. (15 min) 2) Trainees are then provided examples where they must determine the directionality of travel and the relationships being represented. (30 min) 3) Instructor to take up questions and provide feedback. (5 min)	
Instructor Resources		**References**
1) PowerPoint of Strip Plot 2) DWAN or standalone workstation 3) Projector 4) Trainee worksheets 5) Instructor DS solution 6) Maneuvering boards 7) Calculators/per trainee 8) Scrap paper (calculations) 9) Pencils 10) Rolling / parallel rulers		1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook
Note(s) to Instructor		
Nil.		

EO 004.02 – VALIDATE TMA DATA
EC 004.02 – VALIDATE TMA DATA

EC 004.02 – Validate TMA Data		
Scope	**Conditions**	**Instructional Sequence**
EO 004.02	Unclassified Classroom	After LG 004.02.04
Time	**Classification**	**Instructor**
100 min	UNCLASS	RQ-MS SONAR OP

Scenario	EC 004.02 TMA scenario
Assessment Guidance	**Input:** Trainees will be provided with a TMA scenario and the listed resources required to complete a Line of Sound diagram, determine contacts relationships, and provide estimates on contacts course, speed, range and time of CPA. **Performance:** Trainees will complete; Line of Sound diagram; determine a contacts relationship to ownship from the provided scenario, provide an estimate on the contacts course (+/- 30 degrees), speed (+/- 2 knots), range (+/- 2000 yards) and determine the Closest Point of Approach using the provided maneuvering board. **Output:** Trainee must achieve 20/26 SATs to pass this EC. **Feedback:** Immediate verbal and written feedback IAW EC 004.02 checklist upon completion of the EC.

Instructor Resources	References
1) Maneuvering board 2) LOS work sheet 3) Ruler 4) Pencils 5) Scientific calculator 6) Formula sheet	1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook 3) CMS 330 Combat System User Manual (CSUM)

Note(s) to Instructor	
Nil.	
Assessment Limitations	Nil.
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

EC 004.02 VALIDATE TMA DATA			
Item	**SAT**	**UNSAT**	**Remarks**
Complete LOS diagrams to include identifying the following:			
LOS	☐	☐	
TBO	☐	☐	
OAA	☐	☐	
TAA	☐	☐	
OCS	☐	☐	
TCS	☐	☐	
OSI	☐	☐	
TSI	☐	☐	
OSA	☐	☐	
TSA	☐	☐	
RSI	☐	☐	
RSA	☐	☐	
Determine Doppler relationships to include:			
Contact relationship to Ownship	☐	☐	
Ownship relationship to Contact	☐	☐	
Overall Opening/Closing/CPA relationship	☐	☐	
Recognize all contact manoeuvres	☐	☐	
Determine applicable Ranging methods to include:			
Single Eklund (if applicable)	☐	☐	
Double Leg Eklund (if applicable)	☐	☐	
Frequency Rate Eklund (if applicable)	☐	☐	
Warren Ranging (if applicable)	☐	☐	
Recognize Geographic Strip plot patterns to include:			
Zero Bearing Rate Pattern (if applicable)	☐	☐	
Fan Pattern (if applicable)	☐	☐	
Intersecting Pattern (if applicable)	☐	☐	
Conduct Relative Velocity on a maneuvering board to determine the following:			
CPA bearing plot	☐	☐	
Distance to CPA	☐	☐	
Time to CPA	☐	☐	
Score: / 26 **The trainee must achieve 20/26 SATs to pass this assessment.**			

EC 004.02 VALIDATE TMA DATA		
Positive Feedback	**Improvement Feedback**	
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 004 – CONDUCT TMA
PC 004 – CONDUCT TMA

PC 004 – Conduct TMA		
Scope	**Location Conditions**	**Instructional Sequence**
PO 004	SEACOT	After EC 004.01
Time	**Classification**	**Instructor**
300 min	SECRET	RQ-MS SONAR OP

Scenario	IAW PC 004 scenario
Assessment Guidance	**Input:** Trainee will be at a CMS console in SPS or SCS role, provided TMA scenario with a subsurface contact supplied through SEACOT CANTASS, a blank Line of Sound diagram, and the listed resources. **Performance:** Trainee must set-up and operate CMS to run a TMA solution and create an Area Of Probability. Trainee will draw and label the Line of Sound diagram with the information provided in the scenario. Trainee must verbally indicate the contact relationships, the Geographic Strip plot pattern(s), and indicate which ranging method(s) are applicable IAW PC 004 checklist. (20 min per trainee) **Output:** Trainee must achieve 38 / 50 SATs to pass this assessment. **Feedback:** Immediate verbal and written feedback IAW PC 004 checklist. (5 mins per trainee)

Resources	**References**
1) Game piece scenario 2) Go-operator 3) Line of Sound diagram (blank)	1) NCPM 600 2) NOSTP 201 Relative Velocity Handbook 3) CMS 330 Combat System User Manual (CSUM)

Note(s) to Instructor	
Nil.	
Assessment Limitations	Nil.
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

PC 004 CONDUCT TMA			
Item	**SAT**	**UNSAT**	**Remarks**
Login under SPSO role	☐	☐	
Create and calculate a TMA TDA solution to include:			
Select Tactical Operations tab	☐	☐	
Select Tactical Decision Aids	☐	☐	
TDA manager select the UWW button	☐	☐	
Select Target Motion Analysis	☐	☐	
Enter Title	☐	☐	
Hook Bearing Line and Select TACPLOT entry	☐	☐	
Select Global TMA configuration	☐	☐	
Input given ranges then Save and Exit	☐	☐	
Verify VOSIW	☐	☐	
Select Ignore CANTASS Track History	☐	☐	
Select Publish AOP and STM	☐	☐	
Select compute	☐	☐	
Publish TMA data to include:			
Confirm TDA is displayed in the TAC right column with the title CCS tracking trial	☐	☐	
Operator record the TMA ID # (i.e., 4993493066)	☐	☐	
Select Edit Solution	☐	☐	
Verify the TMA ID number matches the one recorded above	☐	☐	
Verify CANTASS track and its ambiguous bearing lines are being plotted	☐	☐	
Select Constraints and Seeds to verify contact speed min/max for AOP	☐	☐	
Select the solutions tab to observe the graph being plotted	☐	☐	
Resolve for ambiguity procedures to include:			
After manoeuver verify false bearing eliminated from solution	☐	☐	
Select track with bearing ambiguity (i.e., 1001-2001);	☐	☐	
Select BRG1 or BRG2 (true bearing xxx) at the CANTASS adapter presentation panel	☐	☐	

PC 004 CONDUCT TMA			
Select Picture Compilation	☐	☐	
Select UWW Manual	☐	☐	
Delete Ambiguous Bearing from CTC list	☐	☐	
Record time	☐	☐	
Analyze LOS diagrams to include identifying the following on a maneuvering board:			
LOS	☐	☐	
TBO	☐	☐	
OAA	☐	☐	
TAA	☐	☐	
OCS	☐	☐	
TCS	☐	☐	
OSI	☐	☐	
TSI	☐	☐	
OSA	☐	☐	
TSA	☐	☐	
RSI	☐	☐	
RSA	☐	☐	
Determine Doppler relationships to include:			
Contact relationship to Ownship	☐	☐	
Ownship relationship to Contact	☐	☐	
Overall Opening/Closing/CPA relationship	☐	☐	
Recognize all contact maneuvers	☐	☐	
Determine applicable Ranging methods to include:			
Single Eklund (if applicable)	☐	☐	
Double Leg Eklund (if applicable)	☐	☐	
Frequency Rate Eklund (if applicable)	☐	☐	
Warren Ranging (if applicable)	☐	☐	
Recognize Geographic Strip plot patterns to include:			
Zero Bearing Rate Pattern (if applicable)	☐	☐	
Fan Pattern (if applicable)	☐	☐	
Intersecting Pattern (if applicable)	☐	☐	

Score: /50

The trainee must achieve 38 / 50 SATs to pass this assessment.

PC 004 CONDUCT TMA	
Positive Feedback	**Improvement Feedback**

Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION

1. **Performance**: Brief/disseminate intelligence/oceanographic information (B0040, B0035).

2. **Conditions**:

   a. Given:

      (1) ARP system,
      (2) Bathythermograph data,
      (3) DWAN access,
      (4) METOC oceanographic data,
      (5) ARP flash-up SOP,
      (6) References, and
      (7) Supervision.

   b. Denied: Nil; and

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at sea.

3. **Standard**: The acoustic operator shall Brief/Disseminate Intelligence/Oceanographic Information (B0040, B0035) by:

   a. Interpreting intelligence/oceanographic data (F0045) by:

      (1) Providing QA for bathymetric forms/data (C0046), and
      (2) Analyzing XBT/XSV data (C0040), and

   b. Determining oceanographic effects on acoustic operations (C0150) by:

      (1) Setting up oceanographic equipment (C0012),
      (2) Operating oceanographic equipment (C0030),
      (3) Calculating acoustic sensor range info (C0115), and
      (4) Providing recommendation for optimum sensor placement.

4. **Training Limitations**: Acoustic Range Prediction system at Naval Fleet Schools is Unclassified and does not contain real world acoustic data.

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
LG 005.01.01 – INTERPRET OCEANOGRAPHIC PRODUCTS

LG 005.01.01 – Interpret Oceanographic Products		
Location Conditions	**Instructional Sequence**	
Acoustic Range Prediction Classroom	Prior to PO 003	
Time	**Classification**	**Instructor**
100 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Interpret oceanographic products to include: a. OFA b. Sea Surface Temperature (SST) charts c. Weather reports d. Bathymetric data e. Bottom composition charts (USN Environmental guides)	1) Think/Share 2) Collaborate Study 3) Group Presentation	
Scenario	Nil.	
Activity Steps	1) Instructor to ask class to individually create a list of oceanographic products they believe are important in their role to contribute to the oceanographic brief. Trainees have 3 min to compile a list. (3 min - Method 1) 2) Trainees are then asked to share one of their responses in order to build a class list on the white board (instructor to compile the list as trainees give their responses). Instructor to confirm the list includes all items listed in the learning objectives (2 min - Method 1) 3) Instructor will then break trainees into groups and assign 1-3 of the products (based on class size). Groups are given example products and must create a 3-5 min presentation on the information found in the product, its significance and where to find the products. If trainees do not have sufficient knowledge to support this, learning support handouts are to be given to ensure the content is covered and accurate. (45 min - Method 2) 4) Groups are then to present their content to the class. Instructor to provide feedback and add any missing content to ensure all information is covered to the level required (45 min - Method 3) 5) Instructor to lead a quick close out discussion on the significance of these products and their content. (5 min)	
Instructor Resources	**References**	
1) Instructor DWAN workstation 2) Projector 3) Example oceanographic products to include: a. Ocean Feature Analysis b. SST charts	1) MILOC 2) RP33 3) US Navy Environmental Guides 4) METOC Webpage 5) NOAA Webpage 6) Ocean Navigator Webpage	

LG 005.01.01 – Interpret Oceanographic Products	
c. Weather reports d. Bathymetric data e. Bottom composition chart 4) Learning support handouts on each topic 5) White board and markers 6) Flip chart & markers	
Note(s) to Instructor	
1) Ensure distribution of experience in groups, pay close attention to trainee's ability to conduct the activity. If they are showing signs of frustration or unable to answer the 3 questions, provide the "learning support handouts". 2) Instructor must be fluent in the use of the various webpages required and demonstrate this to trainees.	

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
LG 005.01.02 – SOUND PROPAGATION

LG 005.01.02 – Sound Propagation		
Location Conditions	**Instructional Sequence**	
Acoustic Range Prediction Classroom	After LG 005.01.01	
Time	**Classification**	**Instructor**
150 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Determine variability of ocean thermal structure 2) Determine Sound paths by extrapolating from XBT data	1) Lecture 2) EDI 3) Individual Problem Solving	
Scenario	Nil.	
Activity Steps	1) Instructor to deliver lecture on ocean thermal structure to cover: (50 minutes – Method 1) a. Geographic variability b. Daily variability c. Seasonal variability d. Geophysical variability 2) Instructor to deliver lecture on sound paths to include: (50 minutes – Method 1) a. Reflection/refraction b. Snell's Law/HALT rule c. Cut-off frequency d. 5 basic sound paths 3) Instructor to draw bathy data on the whiteboard and have trainees draw a ray trace based on the bathy data provided. 5 examples to be completed, with different trainee conducting the ray trace each time. (25 minutes – Method 2) 4) Instructor to give trainees cut-off frequency worksheets then do an example with the class. Trainees will individually complete the worksheet. Upon completion, the instructor will have trainees correct	

LG 005.01.02 – Sound Propagation	
	their sheets and provide feedback on any questions that trainees had trouble with. (25 minutes – Method 3)

Instructor Resources	References
1) Instructor DWAN workstation 2) Projector 3) PowerPoint lecture for Ocean Thermal structure 4) PowerPoint lecture for Sound Paths 5) Example data for bathy activity (x5) 6) Cut off frequency worksheet 7) Cut off frequency DS solution	1) MILOC 2) RP 33 3) US Navy Environmental Guides

Note(s) to Instructor
Nil.

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
LG 005.01.03 – EFFECTS OF WATER MASSES ON ASW SENSORS

LG 005.01.03 – Effects of Water Masses on ASW Sensors		
Location Conditions		**Instructional Sequence**
Acoustic Range Prediction Classroom		After LG 005.01.02
Time	**Classification**	**Instructor**
100 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Effects of water masses on sensors		1) Floating Flip Chart 2) Self-Study 3) Lecture
Scenario	Nil.	
Activity Steps	1) Instructor to create 4 flip charts, one for each category: a. Internal waves b. Ocean fronts c. Eddies d. Currents 2) Trainees are each assigned a flip chart and asked to write down everything they know about that topic (paired up if class size exceeds 4 trainees). Trainees will write down everything they know about that topic. (2 minutes – Method 1) 3) Trainees are then directed to move to the next flip chart, review what the last group wrote, and add to it. (3 minutes – Method 1) 4) Trainees will do 2 more rotations until all trainees have had an opportunity to add to each station. (6 minutes – Method 1) 5) Instructor to assign readings from MILOC sections for self-study: (50 minutes – Method 2) a. Internal waves	

LG 005.01.03 – Effects of Water Masses on ASW Sensors	
	b. Ocean fronts, eddies and currents 6) Upon completion of the readings, trainees are then to conduct the floating flip chart activity again and add to the lists based on information from the readings (2 min per chart). (8 minutes – Method 1) 7) Instructor will have trainees present their findings and add anything that may have been missed. (7 minutes – Method 1) 8) Instructor to then conduct lecture on water masses effect on ASW sensors, taking extra time to focus on any areas trainees struggled with during the flip chart activity. (20 minutes – Method 3)
Instructor Resources	**References**
1) Instructor DWAN workstation 2) Projector 3) MILOC manual/per trainee (web based or paper) 4) 4 flip charts and markers 5) PowerPoint lecture for water masses effect on ASW sensors	1) MILOC 2) RP 33 3) METOC Webpage 4) US Navy Environmental Guides
Note(s) to Instructor	
Nil.	

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
LG 005.01.04 – FACTORS AFFECTING ASW OPS IN COASTAL WATERS

LG 005.01.04 – Factors Affecting ASW Ops in Coastal Waters		
Location Conditions		**Instructional Sequence**
Acoustic Range Prediction Classroom		After LG 005.01.03
Time	**Classification**	**Instructor**
100 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Determine effects of: a. Bottom topography & depth b. Sea life c. Shipping d. Fresh water run-off 2) Identify the significance regarding sensor placement and functionality		1) Lecture 2) 3 Star 3) Bottom topography activity
Scenario	Nil.	
Activity Steps	1) Instructor to conduct a lecture on factors affecting ASW ops in coastal ops. (35 min - Method 1) 2) Trainees are then directed to create a list of their "top three" points they took from the lesson. What is the most important or most interesting of the lesson. (2 min - Method 2) 3) Trainees are then to share their responses with the class. (3 min - Method 2) 4) Trainees are paired up and given a bottom topography diagram worksheets which they must analyze and make recommendations for sensor placement, based on instructor designated ship placement. (40 min - Method 3) 5) Instructor to review Wenz Curve with class. (20 min - Method 1)	
Instructor Resources		**References**
1) Instructor Workstation 2) Projector 3) PowerPoint lesson for coastal ops 4) Bottom topography worksheets 5) Wenz curve handout		1) MILOC 2) RP 33 3) USN Environmental Guides 4) METOC website 5) NOAA website
Note(s) to Instructor		
Nil.		

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
LG 005.01.05 – FOM CALCULATIONS

LG 005.01.05 – FOM Calculations		
Location Conditions		**Instructional Sequence**
Acoustic Range Prediction Classroom		After LG 005.01.04
Time	**Classification**	**Instructor**
400 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Calculate Figure of Merit		1) Lecture 2) EDI
Scenario	Nil.	
Activity Steps	1) Instructor will conduct a lesson on calculating FOM and all the required information in order to do so. (100 min - Method 1) 2) Trainees are then given a worksheet where they will complete the FOM calculations with instructor support as required. Instructor to take worksheets up with class and provide feedback. (300 min - Method 2)	
Instructor Resources		**References**
1) PowerPoint on FOM 2) Db math chart 3) Wenz curve 4) FOM worksheets 5) FOM DS solution		1) MILOC 2) RP 33 3) NCPM 600 4) MARLANTORD RN 001/04
Note(s) to Instructor		
Nil.		

EO 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA
EC 005.01 – INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA

EC 005.01 – Interpret Intelligence/Oceanographic Data		
Scope	**Conditions**	**Instructional Sequence**
EO 005.01	Acoustic Range Prediction Classroom	After LG 005.01.05
Time	**Classification**	**Instructor**
300 min		RQ-MS SONAR OP

Scenario	IAW EC 005.01 scenario package
Assessment Guidance	**Input:** Trainees will be provided with a scenario package and EC 004.02 checklist. **Performance:** Trainees will individually interpret and analyze the oceanographic data effects on sensors verbally explain their understanding of effects of oceanographic data on ASW operation. (20 minutes per trainee) **Output:** Trainee must achieve 21/28 SATs to pass this assessment. **Feedback:** Provide verbal and written feedback IAW EC 005.01 checklist. (5 minutes per trainee)

Instructor Resources	**References**
1) Scenario package 2) EC 004.02 checklist 3) Wentz Curve 4) DS solution 5) Db math chart	1) MILOC 2) RP 33 3) NCPM 600 4) MARLANTORD RN 001/04

Note(s) to Instructor	
Out scenario package and checklist night before to allow trainees to review prior to test.	
Assessment Limitations	Nil.
Impact of Failure	Retrain and re-assessment.

PROTECTED B (when complete)

EC 005.01 INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA			
Item	**SAT**	**UNSAT**	**Remarks**
Explain the Ocean Features Analysis (OFA) and their effects on ASW			
Warm water masses	☐	☐	
Cold water masses	☐	☐	
Current/fronts	☐	☐	
Eddies (warm/cold)	☐	☐	
Describe the effects of Bottom Topography and Composition			
Bottom composition and effects	☐	☐	
Shallow water	☐	☐	
Slope water	☐	☐	
Deep water	☐	☐	
Pinnacle/seamounts/ridges/trenches	☐	☐	
Explain the effects of Chemistry of Water in the region			
Temperature	☐	☐	
Salinity	☐	☐	
Pressure	☐	☐	
Biologics	☐	☐	
Explain the Sound Velocity Profile and its meaning on ASW operations			
Upper mixed layer	☐	☐	
Sonic layer depth	☐	☐	
Deep sound channel axis	☐	☐	
Conjugate depth	☐	☐	
Depth access	☐	☐	
Identify the potential Sound Paths present and their effects			
Direct path	☐	☐	
Bottom bounce	☐	☐	
Deep sound channel	☐	☐	
Convergence zone	☐	☐	
Surface duct/half channel	☐	☐	
Using the Full Field display, identify the presence of the following sound characteristics			
Bottom bounce	☐	☐	
Convergence zone	☐	☐	
Annulus	☐	☐	
Deep sound channel	☐	☐	
Shadow zones	☐	☐	

EC 005.01 INTERPRET INTELLIGENCE/OCEANOGRAPHIC DATA		
Score: /28 **The trainee must achieve 21/28 SATs to pass this assessment.**		
Positive Feedback	**Improvement Feedback**	
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
LG 005.02.01 – SETTING UP OCEANOGRAPHIC EQUIPMENT

LG 005.02.01 – Setting up oceanographic equipment		
Location Conditions		**Instructional Sequence**
Acoustic Range Prediction Classroom		After EC 005.01
Time	**Classification**	**Instructor**
900 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives		**Instructional Method(s)**
1) Set up and configure oceanographic equipment (ARPS) 2) Shutting down ARPS 3) Operating oceanographic equipment (ARPS) by Importing oceanographic data		1) EDI
Scenario	Nil.	
Activity Steps	1) Instructor to explain and demonstrate how to set up ARPS on the projector, including pulling data from ocean navigator. (100 min - Method 1) 2) Instructor to demonstrate how to shut-down the system, trainee to follow along. (100 min - Method 1) 3) Trainees are to then follow along setting up ARPS as the instructor talks them through and demonstrates. (200 min - Method 1) 4) Instructor to explain and demonstrate how to operate ARPS on the projector. (Conduct explain demo of one, then have trainees follow along on their system, then move to next procedure to explain and demo before having trainees follow along on their systems). (500 min - Method 1) a. Ingest data b. View data c. Create a platform/TGT	
Instructor Resources		**References**
1) NITES2R Laptops 1/trainee 2) Projector 3) Instructor NITES2R Console 4) NITES2R SOP 5) Whiteboard 6) DWAN workstation 7) Markers 8) Scenarios		1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage
Note(s) to Instructor		
1) Due to equipment limitations/failures anticipate system crashes and operator frustration throughout. Shut down of system is to be done in follow-on lesson IAW SOP's. Instructor to ensure GIOPS day data is pulled before start of course from DWAN or GPNET.		

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
EC 005.02.01 – SETTING UP / OPERATING OCEANOGRAPHIC EQUIPMENT

EC 005.02 – Setting Up / Operating Oceanographic Equipment		
Scope	**Conditions**	**Instructional Sequence**
EO 005.02	Acoustic Range Prediction Classroom	After LG 005.02.01
Time	**Classification**	**Instructor**
300 min	UNCLASS	RQ-MS SONAR OP x 2
Scenario	N/A	
Assessment Guidance	**Input:** Trainees will be provided with an ARPS and conduct the set up procedures. **Performance:** Trainees will flash-up, setup and shut-down the ARPS. (45 mins per trainee with 2 instructors) **Output:** Trainee must achieve 20/24 SATs to pass this assessment. **Feedback:** Provide verbal and written feedback IAW EC 005.02 checklist. (5 mins per trainee with 2 instructors)	
Instructor Resources	**References**	
1) NITES2R Laptops 1/trainee 2) NITES2R SOP	1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage	
Note(s) to Instructor		
1) An additional RQ-MS SONAR Op instructor is required if there are more than 6 trainees as per Chapter 1 – Required Course Staffing.		
Assessment Limitations	ARP system may have technical issues that impede trainee's ability to complete all checklist items. Instructor may need to move to another ARPS computer or simulate the process.	
Impact of Failure	Retrain and re-assessment.	

PROTECTED B (when complete)

EC 005.02.01 SETUP ACOUSTIC RANGE PREDICTION SYSTEM			
Item	**SAT**	**UNSAT**	**Remarks**
Flash-up ARPS			
Start VMWare Workstation	☐	☐	
Navigate between virtual machines	☐	☐	
Use full screen mode	☐	☐	
Start the N2R edge node (LOCEDGGWY)	☐	☐	
Start the N2R client (NITES2R)	☐	☐	
Configure ARPS			
Select area of interest	☐	☐	
Set current DTG	☐	☐	
Setting selected DTG	☐	☐	
Setting change the start date	☐	☐	
Create mission parameters	☐	☐	
Create data orders	☐	☐	
Place a data order	☐	☐	
Import Data Into ARPS			
Decode BT Messages	☐	☐	
Perform manual BT data entry	☐	☐	
Ingest BT message into NITES2R	☐	☐	
Edit BT data values	☐	☐	
View saved BT entries	☐	☐	
Ingest GIOPS NetCDF data in N2R	☐	☐	
Verify ingested GIOPS data	☐	☐	
Calculate derived 2-D products	☐	☐	
Conduct environment analysis with GIOPS forecast data	☐	☐	
Shut Down ARPS			
Perform N2R Client (NITES2R) shutdown procedure	☐	☐	
Perform N2R edge node (LOCEDGGWY) shutdown procedure	☐	☐	
Close VMWare workstation and shutdown the host OS	☐	☐	
Score: /24 **The trainee must achieve 20/24 SATs to pass this assessment.**			

EC 005.02.01 SETUP ACOUSTIC RANGE PREDICTION SYSTEM		
Positive Feedback	**Improvement Feedback**	
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
LG 005.02.02 – CALCULATING ACOUSTIC SENSOR RANGE INFORMATION

LG 005.02.02 – Calculating acoustic sensor range information		
Location Conditions	**Instructional Sequence**	
Acoustic Range Prediction Classroom	After EC 005.02.01	
Time	**Classification**	**Instructor**
1200 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Determine oceanographic effects on acoustic ops 2) Calculate acoustic sonar range info	1) EDI 2) Practical	
Scenario	Nil.	
Activity Steps	1) Instructor will demonstrate how to see and interpret the oceanographic data from ARPS. (50 min - Method 1) 2) Trainees are to then follow along operating ARPS as the instructor talks. Instructor to ensure emphasis is put on how to determine the oceanographic effects in calculating the ranges. (350 min - Method 1) 3) Trainees are given scenarios where they will practice setting up and operating ARPS to calculate acoustic sonar ranges. (800 min - Method 2)	
Instructor Resources	**References**	
1) NITES2R Laptops 1/trainee 2) Projector 3) Instructor NITES2R Console 4) NITES2R SOP 5) Whiteboard 6) Instructor DWAN workstation 7) Markers 8) Scenarios	1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage	
Note(s) to Instructor		
1) Due to equipment limitations/failures anticipate system crashes and operator frustration throughout. Instructor to ensure GIOPS day data is pulled before start of course from DWAN or GPNET.		

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
LG 005.02.03 – ASW MISSION BRIEF

LG 005.02.03 – ASW Mission Brief		
Location Conditions		**Instructional Sequence**
Classroom		Prior to LG 005.02.04
Time	**Classification**	**Instructor**
100 min	SECRET	RQ-PO1 SONAR OP / Lt(N) NWO UWWD
Learning Objectives		**Instructional Method(s)**
1) Introduction to ASW mission brief 2) Understanding of Oceanographic conditions 3) Introduction to Tactical implications		1) Mission Brief / Lecture
Scenario	Operational Mission Scenario IAW NWTP Vol 4.	
Activity Steps	1) Instructor will present an ASW mission brief to the trainees using either an ARBITER/VARANGIAN or other scenarios. (40 mins - Method 1) 2) Instructor to emphasize the importance of understanding the Oceanographic conditions to include: (20 mins - Method 1) a. Bottom topography/composition b. Currents, fronts and eddies c. Sound velocity profile d. Sound paths e. Shadow zones f. Shipping and weather g. Expected SONAR ranges. 3) Instructor to explain how submarines will exploit oceanographic conditions to their tactical advantages and how we would employ sensors to mitigate this advantage. (20 mins - Method 1) 4) Trainees will be allowed to ask questions at the end of the presentations. (20 mins)	
Instructor Resources		**References**
1) CSNI or secret stand-alone computers 2) Projector 3) whiteboard & markers 4) Operational mission scenario (I.e. ARBITER, VARANGIAN, etc)		1) MILOC 2) RP 33 3) METOC webpage 4) ATP-28
Note(s) to Instructor		
1) Instructor need to emphasize to the trainees that they should strive to achieve this level of tactical understanding during their Oceanographic briefs.		

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
LG 005.02.04 – CREATE OCEANOGRAPHIC BRIEF

LG 005.03.02 – Create Oceanographic Brief		
Location Conditions	**Instructional Sequence**	
Acoustic Range Prediction Classroom	After LG 005.02.03	
Time	**Classification**	**Instructor**
750 min	UNCLASS	RQ-MS SONAR OP
Learning Objectives	**Instructional Method(s)**	
1) Providing recommendation for optimum sensor placement by creating an oceanographic brief	1) Trainees brief preparation 2) Trainee demonstrate presentations	
Scenario	IAW LG 005.02.03 Mission Brief.	
Activity Steps	1) Instructor to provide trainees a template for an oceanographic brief. 2) Trainees are to then build an oceanographic brief utilizing ARPS data slides they have created. (350 min - Method 1) 3) Trainees are provided an opportunity to present their brief to peers and or instructors during the allocated brief building time IOT gain feedback prior to EC. (400 - Method 2)	
Instructor Resources	**References**	
1) NITES2R laptops/per trainee 2) NITES2R instructor workstation 3) Projector 4) DWAN workstation 5) NITES2R SOP 6) FOM worksheets 7) Wenz Curve chart 8) dB math chart 9) Mission scenario IAW LG 005.02.03	1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage 6) NCPM 600 7) MARLANTORD RN 001/04	
Note(s) to Instructor		
Nil.		

EO 005.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS
EC 005.02.02 – DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATIONS

EC 005.02.02 – Determine Oceanographic Effects on Acoustic Operations		
Scope	**Conditions**	**Instructional Sequence**
EO 005.02	Acoustic Range Prediction Classroom	After LG 005.03.04
Time	**Classification**	**Instructor**
400 min	UNCLASS	RQ-MS SONAR OP

Scenario	IAW LG 005.02.03 Mission Brief.
Assessment Guidance	**Input:** Trainees will be provided a scenario where they must utilize ARPS to calculate ranges and create an oceanography brief. **Performance:** Trainees will calculate ranges, create and present the oceanography brief. Trainee briefs are to be approx. 10 min in length followed by a 5 min question period. **Output:** Trainee must achieve 24 / 29 SATs on non-critical factors to pass this EC**.** **Feedback:** Verbal and written feedback IAW EC 005.02.02 checklist.

Instructor Resources	References
1) NITES2R laptops/per trainee 2) NITES2R instructor workstation 3) Projector 4) DWAN workstation 5) NITES2R SOP 6) FOM worksheets 7) Wenz Curve chart 8) dB math chart 9) Mission scenario IAW LG 005.02.03	1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage 6) NCPM 600 7) MARLANTORD RN 001/04

Note(s) to Instructor	
Trainees are NOT assessed on presentation style or briefing skills. You may provide feedback to provide opportunity for improvement in briefing skills however this shall not impact the scoring within the checklist.	
Assessment Limitations	Nil
Impact of Failure	Retrain and re-assessment.

PROTECTED B (**when complete**)

EC 005.02.02 DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATION			
Item	**SAT**	**UNSAT**	**Remarks**
1. **Introduction/situation to include**			
a. Classification	☐	☐	
b. References	☐	☐	
c. Track/ship's position	☐	☐	
d. OFA	☐	☐	
2. Brief regional features – to include the impact on sound propagation:			
a. Bottom topography	☐	☐	
b. Bottom composition	☐	☐	
c. Local water masses	☐	☐	
3. Brief SVP features to include:			
a. SVP profile shallow and full	☐	☐	
b. Surface temperature	☐	☐	
c. Sonic layer depth	☐	☐	
d. Critical depth	☐	☐	
e. Depth excess	☐	☐	
f. Deep sound channel axis	☐	☐	
g. Subs best depth to avoid detection	☐	☐	
h. Cut-off frequency	☐	☐	
i. Propagation paths	☐	☐	
j. Possibility of convergence zone	☐	☐	
4. Predicted ranges – display and explain to include:			
a. Full field ADR (HMS/Helo ball)	☐	☐	
b. Active Counter Detection	☐	☐	
c. ADR/ACDR graph	☐	☐	
d. Full field Passive detection ranges for TASS	☐	☐	
e. Full field Passive detection for Sonobuoys	☐	☐	
f. Full field Passive Counter Detection	☐	☐	
g. Representation of field for passive ranges	☐	☐	
5. Brief recommendations to include;			
a. XBT/XSV policy;	☐	☐	
b. Active settings (HMS Helo ball); and	☐	☐	
c. Passive settings (Sono's/TAS)	☐	☐	
d. Provide recommendations that are align with findings and conclusions followed with	☐	☐	

EC 005.02.02 DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATION			
a brief overview of a realistic implementation plan.			
e. Applied principles of oceanographic theory and concepts during the brief.	☐	☐	

Score: /29

The trainee must achieve 24/29 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION
PC 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION

PC 005 – Brief/Disseminate Intelligence/Oceanographic Information		
Scope	**Conditions**	**Instructional Sequence**
PO 005	Acoustic Range Prediction Classroom	After EC 005.02.02
Time	**Classification**	**Instructor**
400 min	UNCLASS	RQ-MS SONAR OP
Scenario	IAW LG 005.02.03 Mission Brief.	
Assessment Guidance	**Input:** Trainees will be provided a scenario where they must utilize ARPS to calculate ranges and create an oceanography brief. **Performance:** Trainees will calculate ranges, create and present the oceanography brief. Trainee briefs are to be approx. 10 min in length followed by a 5 min question period. **Output:** Trainee must achieve 24 / 29 SATs on non-critical factors to pass this EC**.** **Feedback:** Verbal and written feedback IAW PC 005 checklist.	

Instructor Resources	References
1) NITES2R laptops/per trainee 2) NITES2R instructor workstation 3) Projector 4) DWAN workstation 5) NITES2R SOP 6) FOM worksheets 7) Wenz Curve chart 8) dB math chart 9) Mission scenario IAW LG 005.02.03	1) NITES2R user manual 2) MILOC 3) RP 33 4) METOC webpage 5) Ocean Navigator webpage 6) NCPM 600 7) MARLANTORD RN 001/04

Note(s) to Instructor
1) Trainees are NOT assessed on presentation style or briefing skills. You may provide feedback to provide opportunity for improvement in briefing skills however this shall not impact the scoring within the checklist.

Assessment Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54.

PROTECTED B (when complete)

PC 005 DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATION			
Item	**SAT**	**UNSAT**	**Remarks**
1. **Introduction/situation to include**			
a. Classification	☐	☐	
b. References	☐	☐	
c. Track/ship's position	☐	☐	
d. OFA	☐	☐	
2. Brief regional features – to include the impact on sound propagation:			
a. Bottom topography	☐	☐	
b. Bottom composition	☐	☐	
c. Local water masses	☐	☐	
3. Brief SVP features to include:			
a. SVP profile shallow and full	☐	☐	
b. Surface temperature	☐	☐	
c. Sonic layer depth	☐	☐	
d. Critical depth	☐	☐	
e. Depth excess	☐	☐	
f. Deep sound channel axis	☐	☐	
g. Subs best depth to avoid detection	☐	☐	
h. Cut-off frequency	☐	☐	
i. Propagation paths	☐	☐	
j. Possibility of convergence zone	☐	☐	
4. Predicted ranges – display and explain to include:			
a. Full field ADR (HMS/Helo ball)	☐	☐	
b. Active Counter Detection	☐	☐	
c. ADR/ACDR graph	☐	☐	
d. Full field Passive detection ranges for TASS	☐	☐	
e. Full field Passive detection for Sonobuoys	☐	☐	
f. Full field Passive Counter Detection	☐	☐	
g. Representation of field for passive ranges	☐	☐	
5. Brief recommendations to include;			
a. XBT/XSV policy;	☐	☐	
b. Active settings (HMS Helo ball); and	☐	☐	
c. Passive settings (Sono's/TAS)	☐	☐	
d. Provide recommendations that are align with findings and conclusions followed with	☐	☐	

PC 005 DETERMINE OCEANOGRAPHIC EFFECTS ON ACOUSTIC OPERATION			
a brief overview of a realistic implementation plan.			
e. Applied principles of oceanographic theory and concepts during the brief.	☐	☐	

Score: /29

The trainee must achieve 24/29 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. All ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

3. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

4. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

5. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

7. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;

   b. Feedback on the effectiveness of training;

   c. Information for the Training Review Board (TRB) (if required); and

   d. Information to assist in the completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

9. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

10. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

11. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

12. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

13. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

14. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A – CAF or other Official Canadian Government Publications		
A1	NCPM 600	UWW Procedural Manual for HALIFAX Class
A2	NAVORD 3470-1	Maritime Acoustic Security Policy
A3	CFCD 113 Vol 1	Maritime Command Jezebel Analysis Manual
A4	CFCD 113 Vol 2	Maritime Command Acoustic Information Guide (AIG)
A5	C-59-B16-000/MB-001	Acoustic Data Recorder Operation and Maintenance Manual (REACH)
A6	C-59-007-006/MB-001	Sonobuoy Reference Guide
A7	CFCD 106	Maritime Tactical Instructions MARTI 2122
A8	NOSTP 201	Relative Velocity Handbook
A9	MILOC	DG METOC Military Oceanography (MILOC)
A10	MARLANTORD RN 01/04	MARLANTORD Research Note 01/2004 Excerpt
A11	CAN-UM-0001	User's Manual (UM) for Naval Integrated Tactical Environmental System (NITES) II REDESIGN (N2R) Version 5.17.2.0, 10 January 2013
A12	NITES2R SOP	Naval Integrated Tactical Environmental System (NITES) II REDESIGN (N2R) SOP Version 0.8, 8 March 2018
B – Official Allied Force and Allied Government Publications		
B1	ATP 28 Vol 1	Allied Anti-Submarine Warfare Manual
B2	ATP 28 Vol 2	Allied Anti-Submarine Warfare Manual Scientific and Supporting Data Supplement
B3	RP 33	Fleet Oceanographic and acoustic reference manual
B4	USN EG	US Navy Environmental Guides
C – Commercial Publications		
C1		Mk 39 (EMATT) Operators Manual
C2		Mk 39 (EMATT) Field Programmability User Guide
C3	CSUM	CMS 330 Combat System User Manual (CSUM)
D – Other Publications		
D1		WHMIS MSDS Sheets for MK 39 EMATT
D2	NAL	National Acoustic Library (NAL) (TRINITY ADAC)
D3	SPF Report	Sonar Performance Report HMC Ship July 2003
D4	METOC Webpage	http://halifax.mil.ca/MCC/Trinity/pages/metoc_mission.html
D5	NOAA Webpage	https://noaa.govhttps://noaa.gov
D6	Ocean Navigator Webpage	http://navigator.oceansdata.ca/

ANNEX B – TASK LIST

This table provides an overview of all tasks considered and where they are trained within the QSP if selected. No train substantiations are in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
A0005	Perform communications circuit functional checks (comm checks)	NT	
A0010	Operate communications equipment	NT	
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	NT	
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	NT	
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	NT	
B0005	Perform pre-sail checks on equipment	NT	
B0010	Plot formations, screens and dispositions	NT	
B0015	Maintain stateboards	T	003
B0025	Adhere to Emission control (EMCON) restrictions	NT	
B0030	Contribute To Intelligence Gathering Operations	NT	
B0035	Brief Intelligence Information	T	005
B0040	Disseminate Intelligence Information	T	005
B0045	Identify Capacities and limitations of Platforms	NT	
B0050	Operate Command System	T	003
B0055	Maintain Operational Logs	T	003
C0012	Set Up Oceanographic Equipment	T	005
C0015	Operate active sonar equipment	NT	
C0020	Operate Passive Acoustic Equipment	T	003
C0025	Operate ancillary acoustic equipment	T	001
C0030	Operate Oceanographic Equipment	T	005
C0035	Launch underwater expendables	T	001
C0040	Analyze XBT/XSV Data	T	005
C0046	Provide Quality Assurance for Bathymetric forms/DATA	T	005
C0050	Prepare Underwater Expendables for Launch	T	001
C0055	Verify underwater expendables inventories	NT	
C0060	Monitor Sonobuoys	T	003
C0065	Analyze Passive Acoustic Data	T	002
C0070	Track Acoustic Contacts	T	003
C0075	Annotate Recording Media	T	003
C0080	Classify Passive Acoustic Data	T	002
C0085	Localize Passive Acoustic Contacts	T	003
C0086	Setup Target Motion Analysis Plot	T	004

C0087	Perform Target Motion Analysis Calculations	T	004
C0090	Plot Target Motion Analysis (TMA) Data	T	004
C0095	Analyze TMA Data	T	004
C0100	Construct Contact Area Of Probability	T	004
C0105	Correlate active/passive/non-acoustic information	NT	
C0115	Calculate Acoustic Sensor Range Information	T	005
C0125	Analyze acoustic tapes/multi-media	NT	
C0135	Perform Noise Rounds	NT	
C0150	Determine Oceanographic Effects on Acoustic Operations	T	005
C0170	Assist with loading/handling of torpedoes	NT	
C0175	Operate Torpedo Setting Panel (TSP)	NT	
C0180	Fire torpedoes in emergency mode	NT	
C0185	Charge torpedo tubes/flask	NT	
C0195	Perform Passive Aural Listening (PAL)	T	002
C0215	Operate Sonobuoy Processing System (SPS)	T	003
C0255	Conduct TMA	T	004
D0010	Perform UWW weapon systems serviceability checks (i.e. airshot)	NT	
D0025	Perform UWW weapon firings	NT	
D0030	Perform weapons system misfire procedures	NT	
D0080	Analyze LOFAR Grams	T	002
E0020	Amplify Track Information	T	003
F0005	Evaluate correlated acoustic information	NT	
F0010	Record acoustic intelligence data	T	003
F0015	Identify changes in contact's readiness and tactical states	T	004
F0035	Perform basic passive acoustic analysis	T	002
F0045	Interpret Oceanographic Data	T	005
Added Tasks			
	Conduct Preparations for Launching Expendables	T	001
	Create Run Geometry	T	001
	Upload Run Geometry into PTP	T	001

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Dates: 27 January 2021

Location: Rm 322, S15 CFB Stadacona and Virtual via O365

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Bagshaw	J	LCdr (RN)		NTDC(A)
Tallman	J	CPO2	00324	NTDC(A)
Morris	KSA	Lt(N)	00211	NTDC(A)
Craig	B	MS	00324	NFS(A)
Musseau	R	Civ		TRINITY ADAC

Working Group Objective:

1. The objective of this working group was to review the Post Course Report (PCR) and instructor feedback on the RQ-S1 QSP for Session 0004 and 0005.

2. The primary focus was specifically on the indications in the PCR that indicated insufficient Actual Training Time allocated for PO 002 Analyze Passive Acoustic Data and EO 002.01 Perform Passive Aural Listening.

3. The secondary focus was to review checklist items in PC 003 and EC 005.02.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
A0005	Perform communications circuit functional checks (comm checks)	This task is instructed on RQ-S3 course, no requirement for additional training.
A0010	Operate communications equipment	This task is instructed on RQ-S3 course, no requirement for additional training.
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	This task is instructed on RQ-S3 course, no requirement for additional training.
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	No requirement for SONAR Op RQ-S1 to monitor or communicate externally until RQ-PO2 level.
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	No requirement for RQ-S1 level to have training as they do no utilize external communications. This is instructed at the RQ PO1 and RQ PO2 levels.

B0005	Perform pre-sail checks on equipment	This task is instructed on RQ-S3 course, no requirement for additional training.
B0010	Plot formations, screens and dispositions	RQ-S1 position does not have the ability to perform this task on the current equipment.
B0025	Adhere to Emission control (EMCON) restrictions	This task is instructed on RQ-S3 course, no requirement for additional training.
B0030	Contribute to Intelligence Gathering Operations	This task is instructed on RQ-S3 course, no requirement for additional training at this RQ level.
B0045	Identify Capacities and limitations of Platforms	This task is instructed on RQ-S3 course, no requirement for additional training at this RQ level.
C0015	Operate active sonar equipment	This task is instructed on RQ-S3 course, no requirement for additional training.
C0035	Launch underwater expendables	This task is instructed on RQ-S3 course, no requirement for additional training.
C0055	Verify underwater expendables inventories	This is a very simple task of asking RQ levels to count the number of expendables held in inventory. No requirement to teach how to count.
C0105	Correlate active/passive/non-acoustic information	RQ PO2 and RQ PO1 levels are the only positions that have CS MFW's that provide the full tactical picture with the ability to Correlate active/passive/non-acoustic information. No requirement to train this task at this RQ level.
C0125	Analyze acoustic tapes/multi-media	This task is unit specific (ADAC) as it involves post analysis not generally conducted onboard ship.
C0135	Performing noise rounds	This task was already trained in RQ-S3 at the exact same standard.
C0170	Assist with loading/handling of torpedoes	This task is instructed on RQ-S3 course, no requirement for additional training.
C0175	Operate Torpedo Setting Panel (TSP)	This task is instructed on RQ-S3 course, no requirement for additional training.
C0180	Fire torpedoes in emergency mode	This task is instructed on RQ-S3 course, no requirement for additional training.
C0185	Charge torpedo tubes/flask	This task is instructed on RQ-S3 course, no requirement for additional training.
D0010	Perform UWW weapon systems serviceability checks (i.e. airshot)	This task is instructed on RQ-S3 course, no requirement for additional training.
D0025	Perform UWW weapon firings	This task is instructed on RQ-S3 course, no requirement for additional training.

D0030	Perform weapons system misfire procedures	This task is instructed on RQ-S3 course, no requirement for additional training.
F0005	Evaluate correlated acoustic information	This RQ-S1 level does not have the ability to perform this task based on their current equipment.

Discussions / Decisions:

4. EO 002.01 Perform Passive Aural Listening was determined to have insufficient Actual Training Time (ATT) allocated (currently 10 periods of training with 3 periods for the EC). After discussion, an increase of 12 x 50 min periods was agreed upon bringing the ATT to 22 periods of instruction and 3 EC periods.

5. ADAC brought forward concerns on the EC 002.01 standard of 75% as it's significantly higher than the current Fleet Standard of 50%. Concerns were expressed on lowering the standard as SONAR Op submariners still need to achieve the higher standard. As SONAR Op submariners receive more training in AILO, all WG members agreed that the EC standard could to be lowered from 75% but still above the current fleet standard of 50%. The consensus amongst the board members was that 60% was both achievable in the ATT allotted and still reach the standard required.

6. On review of PO 002, para 2(b) updated to nil and 2(c) amended to include environmental conditions in-line with job realities.

7. NFS(P) requested to have group work incorporated into EO 002.02 to help trainees practice identifying characteristics and contacts. ADAC seconded this request. Activities added to several lessons to aid with recall from previous lessons and group work at the end of lessons.

8. PO 002 Analyze Passive Acoustic Data was determined to have insufficient time allotted for TRINITY ADAC to instruct new platforms. It was also determined that additional scenario based training would be beneficial for Acoustic Operators as it fell in line with the Performance Based training methodology. An additional 14 x 50 min periods of instructional training time was allocated, along with some of the ATT re-allocated practical labs to instruction. A new group/class scenario based training was developed in EO 002.02.09 to allow for a team performance activity. An additional 2 x 50 min periods were added to EC 002.02 and PC 002 to allow for instructors to conduct debrief of the EC and PC. This increased the overall time in PO 002 from 77 periods to 105 periods (approximately 3.5 training days).

9. Discussed with Submariner SONAR Op MOC and NTDC(A) Submarine Division CPO2's the modularization requirements of the RQ-SONAR OP-S1 QSP. It was determined from the Military Employment Structure Change Request (MESCR) 4877-4975-1 (CSF SUS/RDIMS 1098556) submitted by Canadian Submarine Force in 27 Oct 2020 that indicates the Submarine SONAR Ops will be required to complete both modules of the RQ-SONAR OP-S1 NQUAL AKYV course to receive the criteria for promotion to Sailor First Class (S1) and subsequent Specialist Level 1 Pay. As a result of this, the QSP will be amended to remove the mention of Submarine SONAR Ops and change to SONAR Operators.

10. After discussion with both the NTDC(A) TDO, NPTG HQ TDO, and Training Manager it was determined that the QSP can remain in 2 Modules with 2 MITE codes (currently 122046 & 122047).

11. There was discussion between the NPTG HQ SONAR OP TM CPO2 Gougeon, NTDC(A) Subs CPO2 Smith and NTDC(A) UWW CPO2 Tallman to determine the AKYF prerequisites. Prerequisites presently indicate the completion of the RQ-S2 OJPR which may not be required for future courses should the above mentioned MESCR be accepted by DPGR. CPO2 Smith had concerns regarding the AILO and AILF becoming prerequisites due to the length of time the accompanying OJPRs take. Version (v) 1.8 of the AKYF QSP had the AILO as a prerequisite which was subsequently dropped in v1.9. Based on discussions and research into historical AKYF QSPs, current AILO and current AILF QSPs, the prerequisites defined in Chapter 1 have been developed.

12. Following completion of the WG while follow-up work was ongoing, NFS(P) brought forward an issue with EC 005.02 – Setup Acoustic Range Prediction System checklist item "Ingest BT Message using LOCEDGGWY". The issue was that the NFS(P) NITES2R systems observed system errors when this method of ingesting BT data into NITES2R was employed, they had found a different method for ingesting BT data and recommended the EC checklist be amended. NTDC(A) staff had an SME test the current method at the NFS(A) NITES2R training and determined that this procedure did work. It was determined that there may differences between the 2 NFS systems and it was determined that as long as the trainees could demonstrate a method of ingesting the BT data into NITES2R then that achieved the training performance goal. The QSP EC 005.02 checklist item has been amended to "Ingest BT Message into NITES2R", thus allowing either method to be utilized for the task performance.

13. In addition, both NFSs indicated a requirement for a SPS Setup & Operation Job Sheet resource for EO 003.02 Operate SPS. NTDC(A) will develop a SPS Setup & Operation Job Sheet resource to be included in future RQ-S1 QSP course packages.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Obtain additional training laptops with WIN RGA program loaded on them	NTDC(A)	TBD
2	Obtain additional PTP for NFS A & P	NTDC(A)	TBD
3	Obtain external packaging for EMATT	NTDC(A)	TBD
4	Training video for Acoustic Data Recorder	NTDC(A)	TBD
5	TMA contact run profiles	NTDC(A)	TBD
6	Acoustic Range Prediction scenarios	NTDC(A)	TBD
7	SPS Setup & Operation Job Sheet	NTDC(A)	TBD

Course History:

14. Ordered oldest to newest of previous QSPs Record of Decision, Record of Proceedings, and Course History annexes. All approval dates are in bold.

Original: **12 September 1997** MS TRG DET: 4985-271 (OPS) (Ver: 1)
Original: **12 September 1997** QSP updated to version two (2) template.
Original: **06 January 2000** QSP update to revision two (2).
Original: **13 February 2002** QSP update to revision three (3).

P-2 Amendment: Dated **20 September 2006**
After a meeting held on the 25 august, 2006, The Sonar Op STDS section amended the PC 403, which included the removal of the low-grade codes/NUCO as they were adequately assessed in EC 403.2. The period removed from this PC has been reallocated as a developmental period.

P-3 Amendment: Dated **02 February 2007**
During meeting between the CFNOS Oceanography Instructor and STDS personnel, it was decided that the current checklist for PC 402 was inadequate and did not challenge the trainee's practical application of Oceanography during a brief to command. The PC checklist was amended for PC 402 in order to better assess the trainee's knowledge of Oceanography IAW the JBOS.

P-4 Amendment: Dated **15 June 2009**
The CF's adaptation of the new version of the MK 39 AAT EMATT and WIN RGA program required an update of the PC 404 check list due to an increase in programmable settings. The PC 405 check list was modified to include internal voice reports, anti-torpedo procedures, and basic tracking of a torpedo and sub-surface contact in order to become more in-line with TDT training.

P-1.4 Amendment: Dated **05 November 2010**
QSP – SONAR OP – QL5A LAST DATE AMENDED: 05 23 2012
After review of the QSP, a revision of provisional numbers was corrected from P-1 through P-5 to read P-1.0 through P-1.4 Standards corrected qualification code on cover page from AIJE to AIXI. 1.6 days was added to course duration. .5 days to PO 401, .5 days to PO 402 and .6 days for Training Administration Day (TAT) resulting in new course duration of 66.7 days.

P- 1.5 Amendment: Dated **23 May 2012**
Administrative amendments to reflect the Command name change and the establishment of new lines of authority.
P-1.6 – Current Update to reflect occupational MESIP requirements. **13 June 2013/23 July 2013/7 November 2016** (assumed versions as not previously noted in Course History).
HCM Hybrid QSP Amendments

These changes are designed to make this QSP HCM Hybrid compliant and ensure capabilities and limitations of new HCM driven equipment are covered. The changes were fully discussed and drafted during the SONAROP HCM QSP working group held 29 Apr – 30 May 13. In attendance were:

PO1 Couture (CFNOS Procedures instructor)
PO1 Jenkins (Senior SONAROP Advisor Future Systems Training Development (FSTD)
PO1 MacDowall (CFNOS Senior SONAROP Quality Assurance Rep)
PO1 Stewart (CFNOS SONAROP Senior Instructor)
PO2 Fisher (CFNOS SONAROP Quality Assurance Rep)

The HCM Hybrid content that will affect this QSP is contained in Annex L – HCM Considerations.

Working Group Amendments – Dated **3 February 2017***
*assumed unapproved version as prior CNPTG approval was 7 November 2016 and next subsequent approval was 5 July 2019 by NTDC(A).

1. The group members thought a modularization of content will enable easier recourse making training more efficient in the same process. The order of the modules was selected to enable the submariners to follow the same course by presenting the content needed by both trades at the beginning.
2. PO 002 Monitor Shipboard Noise should be removed completely since this content was cover fully in the Sonar Op RQ-S3.
3. PO 003's tasks should be reorganised into two Enabling Objectives. That way, the evaluation process will be less time consuming. It also will make it easier to follow the scheduling that the Acoustic data analysis centre (ADAC) already has in place.
4. PO 004's tasks, related to the recorder, should be removed from training since that equipment is not used anymore. Once the new recording equipment is in place a board should be convened to amend training accordingly.
5. Tasks related to PO 005 (TMA) that involves plotting the different types of plots and some calculations should be removed from this course. As TMA calculations are completed through the CMS. Members have determined that Relationships and RELVEL teaching points should remain since they are still relevant to the RQ-S1.
6. PO 006 related to oceanography should be reorganized in order to make evaluations more realistic and closer to the job performance. By doing so, some Enabling Checks will disappear, therefore course will be shortened. However, instructional time for this PO should stay unchanged since those concepts have been difficult to integrate by the trainees.

7. Mine warfare should be introduced in the RQ-S1 course. Sonar Operators have to brief command on the capabilities and limitations of the various types of mines and the threat to HMCS ships. Mine warfare content is not currently covered until the QL6A; it was determined by the working group to be way too late as it is required for S1 to be the unit's SME on mine warfare organization and threats. Therefore, a basic introduction to mine warfare will be integrated to the course to include the various types of threat, with their capabilities and limitations and the mine warfare organization onboard HCMS ships.

Dates: 6-17 May 2019 (approved NTDC(A) **5 July 2019** & CNPTG **9 August 2019**)

Location: Rm 214, S15 CFB Stadacona

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Tallman	J	CPO2	00324	NTDC(A)
Coit	R	PO1	00324	NTDC(A)
Rogers	T	PO2	00324	NTDC(A)
Grandmaison	M	MS	00324	NTDC(A)
Wink	R	MS	00324	NTDC(A)
Willits	S	S1	00324	FDU(A)
Horricks	T	S1	00324	ADAC
Jawornicki	B	SLt	00211	NTDC(A)

Working Group Objective:

1. The objective of this working group was to conduct the changes/amendments identified from the Combat Training Review (CTR). The QSP that was reviewed through the CTR was in Pilot and not yet approved by NPTG(HQ), therefore the working group used this opportunity to switch to the FireFly format while conducting the changes.

Discussions / Decisions:

2. PO 001 Conduct Preparations for Launching Expendables (EMATT program) was discussed and determined that this training will remain the same, only issue is the requirement for additional training assets (sufficient laptops with WIN RGA program, additional Portable Target Programmer (PTP), and external packaging for the EMATT).

3. PO 002 Analyze Passive Acoustic Data was determined to have insufficient time allotted for TRINITY ADAC to instruct new platforms. Additional time was allocated bringing the estimated number of training days to 10 vice the previous 7.

4. PO 003 Operate Command System (SPS) was determined to have outdated content based on the old CCS version of the SPS. Course content was updated to reflect the CMS 330 version of SPS along with subsequent Jobs Sheet Handouts and EC/PC checklists. Additional training was added for the new Acoustic Data Recorder (REACH), along with additional training time allocated for SPS operator Torpedo Detection Training.

5. PO 004 Conduct Target Motion Analysis (TMA) drastically updated. Previous version covered theory and manual TMA plotting, however updated QSP still has TMA theory instructed but plotting is now conducted on the CMS 330 system in SPS mode.

6. PO 005 Brief/Disseminate intelligence/oceanographic information using Acoustic Range Prediction Systems (ARPS) (presently NITES2R) content was acceptable, however the final PC was based on briefing Mine Warfare vice Oceanographic Range predications. This Performance Objective was updated to reflect the task within the MTL and appropriate checklists were created.

Guidance to Development and/or Instructional Staff:

7. NTDCA will continue to update training content and scenarios to provide more dynamic performance based training for the RQ S1 course.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Obtain additional training laptops with WIN RGA program loaded on them	PO1 Coit	
2	Obtain additional PTP for NFS A & P	PO1 Coit	
3	Obtain external packaging for EMATT	PO2 Rogers	
4	Training video for Acoustic Data Recorder	PO1 Coit	
5	TMA contact run profiles	PO2 Rogers	
6	Acoustic Range Prediction scenarios	MS Grandmaison	

Dates: 22 Oct 20 (approved **17 November 2020**).

QSP Amendment

1. IAW CANFORGEN 112/20 RCN JR Rank Designation Change 041200Z SEP 20, all previous RCN JR Rank designations in this QSP have been amended to reflect the new policy.

2. Amendments made to the lesson Enabling Objective naming convention. Changed from Number with Letter to just number. This was to align with NFS Scheduling program requirements.

3. Addition of previously omitted Chapter 3: Assessment of Trainee.

V3.0 Major Amendment CNPTG Approved 8 January 2022

BACKGROUND

1. NTDC(A) received feedback on the RQ-S1 courses Session 0004 conducted at NFS(P) and Session 005 being conducted at NFS(A) that both indicated insufficient training time was allocated for PO 002 – Analyze Acoustic Data EO 002.01 - Perform Passive Aural Listening (PAL) to achieve the standard indicated in EC 002.01. This information was corroborated with TRINITY ADAC, who are the fleet Subject Matter Experts for this training material. Based on this information, the UWW section conducted a half day QSP Working Group on the 27 January 2021 with representation from NTDC(A), NFS(A & P) and TRINITY ADAC with the focus on reviewing EO/EC 002.01.

DISCUSSION

2. During the QSP Working Group, it was determined that both the Actual Training Time was not sufficient for EO 002.01 and the expected standard of the Assessment Plan for EC 002.01 was too high. For the EO 002.01, an additional 12 x 50 minute periods was added to enable sufficient student practice and development of the PAL skills.

3. There was also discussion that the current pass standard for EC 002.01 was 75% when the expected Fleet Standard for all SONAR Operators was only 50%. It was determined that the students needed to achieve a higher standard than the fleet for EC 002.01 but not the current standard of 75%, the consensus was agreed to place the new pass standard for EC 002.01 at 60%.

4. The WG also reviewed the remainder PO 002 – Analyze Passive Acoustic Data and determined that additional Actual Training Time was required to facilitate the instruction of new Acoustic Intelligence and to allow the conduct of a group performance based training scenario. After the review, an additional 15 periods were allocated to EO 002.02 Perform Basic Passive Acoustic Analysis and 1 additional period for PC 002 to allow for instructor to student debrief.

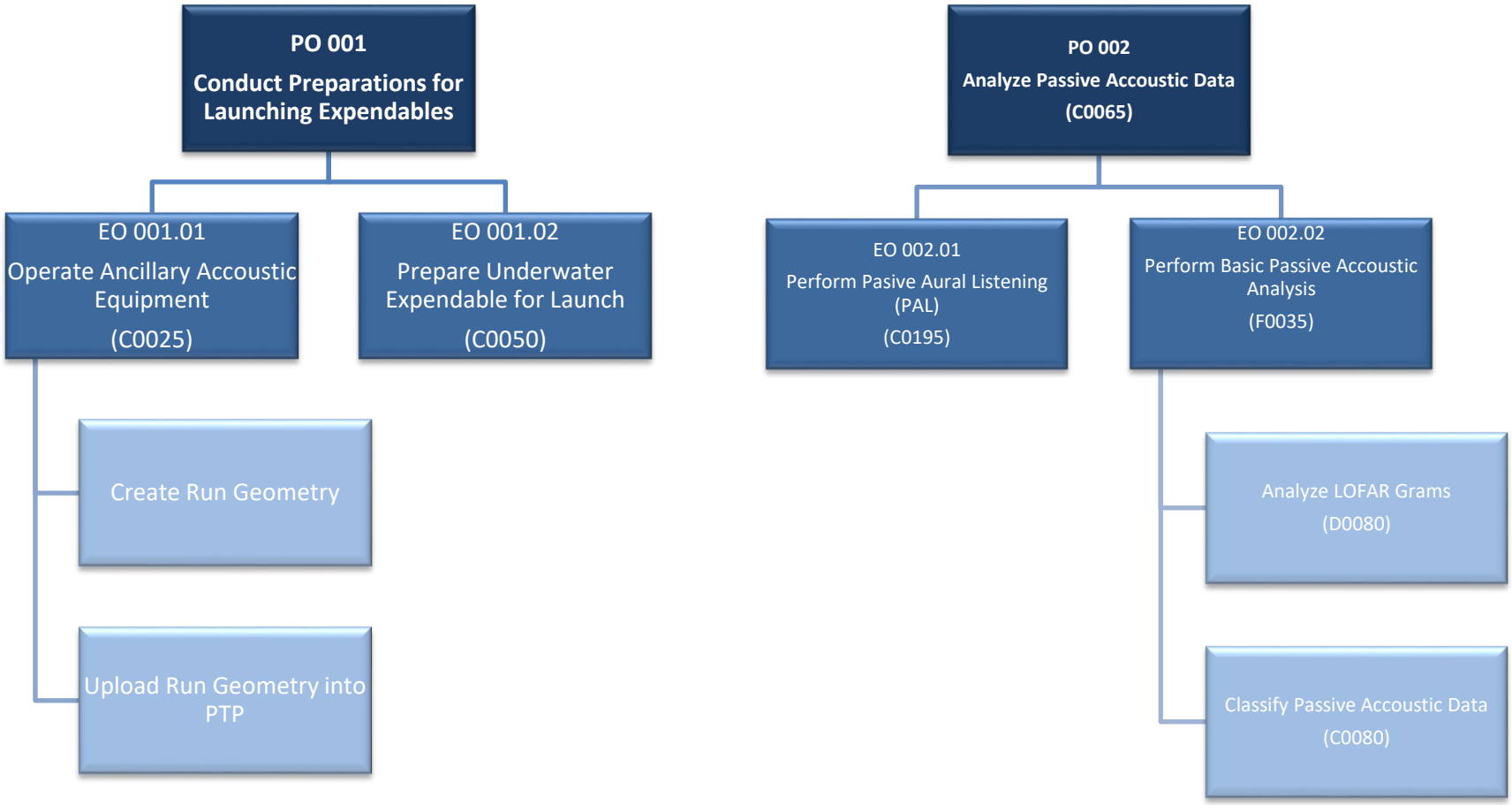
5. The WG also reviewed 2 additional feedback requests; one for review of EC 005.02 Setup Acoustic Range Prediction System (ARPS) checklist specifically as it applied to importing Bathymetric Data into NITES2R, and two, a request to develop an SPS Job Sheet for EO 003.02 Operate Passive Acoustic Equipment (SPS). NTDC(A) UWW Section reviewed the EC 005.02 Checklist and made one amendment to EC checklist item 3.c. to reflect a generic method for importing data as there were multiple methods to perform this task. UWW section will have an EO 003.02 Operate Passive Acoustic Equipment (SPS) Job Sheet aid developed for the next scheduled course.

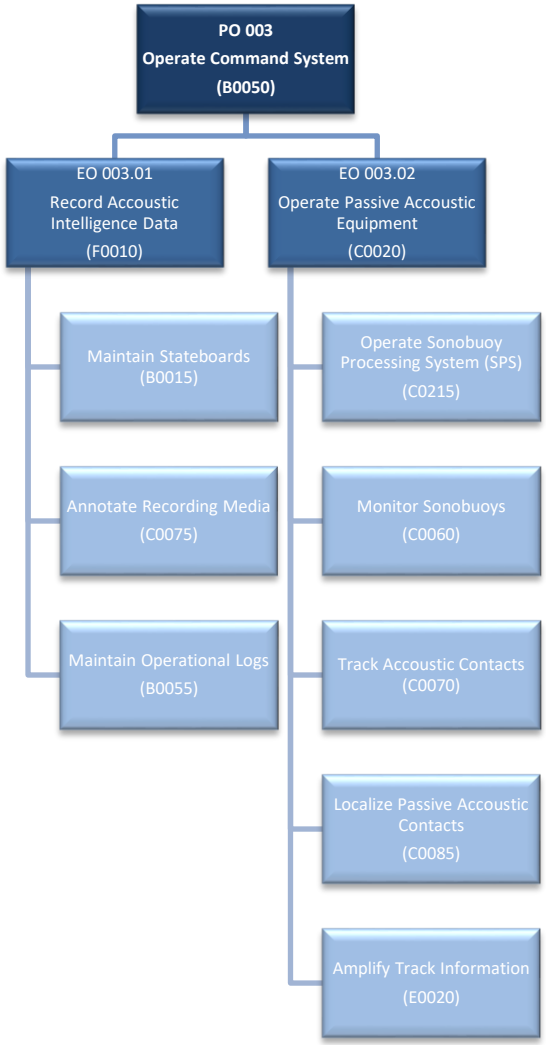
6. NTDC(A) UWW section updated the QSP into a new and approved QSP format with all subsequent Annex's included.

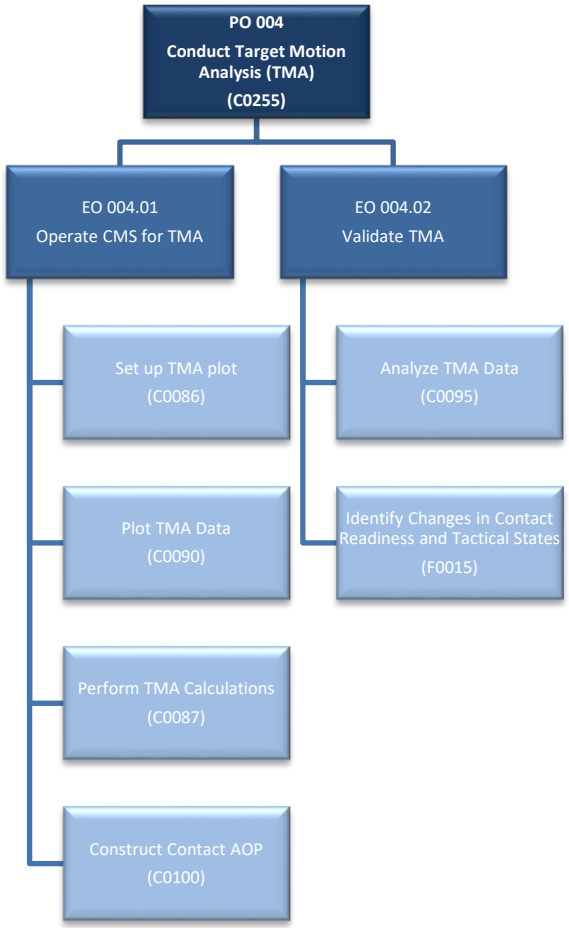
7. Upon completion of the QSP amendments, a full review of the QSP was conducted with update to format, spelling and grammar was conducted to ensure it meets a high standards, and an additional 28 periods of Actual Training Time was allocated for PO 002 along with the inclusion of a new performance based training scenario. This additional training time, along with the required BAT/TAT and PT time has extended the course by a total of 33 periods or approximately 5 calendar days.

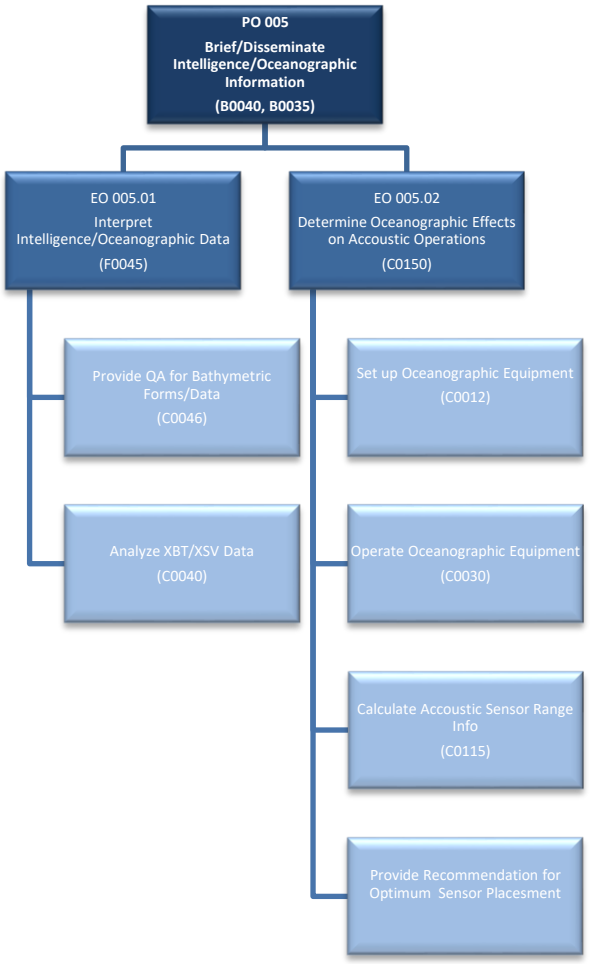
ANNEX D – TASK ANALYSIS DIAGRAM

These diagrams represent the recreation or original task analysis (scalar) used to define the job.




PO 003
Operate Command System
(B0050)
EO 003.01
Record Accoustic Intelligence Data
(F0010)
EO 003.02
Operate Passive Accoustic Equipment
(C0020)
Maintain Stateboards
(B0015)
Annotate Recording Media
(C0075)
Maintain Operational Logs
(B0055)
Operate Sonobuoy Processing System (SPS)
(C0215)
Monitor Sonobuoys
(C0060)
Track Accoustic Contacts
(C0070)
Localize Passive Accoustic Contacts
(C0085)
Amplify Track Information
(E0020)


PO 004
Conduct Target Motion Analysis (TMA)
(C0255)
EO 004.01
Operate CMS for TMA
EO 004.02
Validate TMA
Set up TMA plot
(C0086)
Plot TMA Data
(C0090)
Perform TMA Calculations
(C0087)
Construct Contact AOP
(C0100)
Analyze TMA Data
(C0095)
Identify Changes in Contact Readiness and Tactical States
(F0015)


PO 005
Brief/Disseminate Intelligence/Oceanographic Information
(B0040, B0035)
EO 005.01
Interpret Intelligence/Oceanographic Data
(F0045)
EO 005.02
Determine Oceanographic Effects on Accoustic Operations
(C0150)
Provide QA for Bathymetric Forms/Data
(C0046)
Analyze XBT/XSV Data
(C0040)
Set up Oceanographic Equipment
(C0012)
Operate Oceanographic Equipment
(C0030)
Calculate Accoustic Sensor Range Info
(C0115)
Provide Recommendation for Optimum Sensor Placesment

ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	# Trainees + Instructors /	Unit Price	# Days	Initial Course	Each Successive Course	5 Year Total
Airfare (Return)	8	$1,000.00	n/a	$8,000	$8000	$40,000.00
R&Q	8	$36.00	102	$29,376	$29,376	$146,880.00
Incidentals	8	$17.50	102	$14,280	$14,280	$71,400.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	16	$50.00	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.000	15			
Computer requirements	16	$1,000.00	1	$16,000	$0	$16,000.00
IT Equip (Smart board, etc.)	1	$6,000.00	1	$6,000	$0	$6,000.00
DWAN connection (Setup)	1	$5,000.00	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.00	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.00	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PO1	#	$###.##	###	$8,920	$8,920	$44,600.00
Civilian Instructor						
Subtotal				**$8,920**	**$8,920**	**$44,600.00**
Total Estimated Costs				**$66,165**	**$39,165**	**$336,205**

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
PC 001	PO 001	To assess trainee's competence in conducting preparations for launching expendables.	Practical	Trainee must obtain 33 /41 SATs IAW PC 001 checklist.
EC 002.01	EO 002.01	Perform Passive Aural Listening	Practical	60% on Analysis
EC 002.02	EO 002.02	Perform basic passive acoustic analysis	Practical	75% on Analysis 80% on Classification
PC 002	PO 002	Analyze passive acoustic data	Practical	75% on Analysis 80% on Classification
EC 003.01	EO 003.01	Record acoustic intelligence data.	Practical	Trainee must obtain 8 / 10 SATs IAW EC 003.01 Checklist
EC 003.02.01	EO 003.02	Operate passive acoustic equipment.	Practical	Trainee must obtain 20 / 25 SATs IAW EC 003.02.01 Checklist
EC 003.02.02	EO 003.02	Operate SPS for Anti-Torpedo Procedure	Practical	Trainee must obtain 8 / 10 SATs IAW EC 003.02.02 Checklist
PC 003	PO 003	Operate command system	Practical	Trainee must obtain 24 / 32 SATs IAW PC 003 Checklist
EC 004.01	EO 004.01	Operate CMS for TMA	Practical	Trainee must obtain 17 / 23 SATs IAW EC 004.01 Checklist
EC 004.02	EO 004.02	Validate TMA data	Practical	Trainee must obtain 18 / 24 SATs IAW EC 004.02 Checklist
PC 004	PO 004	Conduct Target Motion Analysis	Practical	Trainee must obtain 14 / 18 SATs IAW PC 004 Checklist

EC 005.01	EO 005.01	Interpret intelligence/oceanographic data	Practical	Trainee must obtain 21 / 28 SATs IAW EC 005.01 Checklist
EC 005.02.01	EO 005.02.01	Setup Acoustic Range Prediction System	Practical	Trainee must obtain 20 / 24 SATs IAW EC 005.02 Checklist
EC 005.02.02	EO 005.02.02	Determine oceanographic effect on acoustic ops	Practical	Trainee must obtain 24 / 29 SATs IAW EC 005.02.02 Checklist
PC 005	PO 005	Brief/Disseminate intelligence/oceanographic information	Practical	Trainee must obtain 24 / 29 SATs IAW PC 005 Checklist

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

This table reflects areas that were identified to have methodology that was not enabling trainees to grasp the Learning Objectives (LOs). Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #
Acoustic Data Recorder	Video needs to be redone	LG 003.01.01	LSR 28-21
Program EMATT Demonstration	Require a demonstration video on how to program a Mk39 EMATT	LG 001.01.01 LG 001.02.01	LSR 27-21 LSR
Submarine Training Shapes	Require 3D models of different submarines to demonstrate hull resonance sound sources	All LG in EO 002.02	LSR 11-21 LSR 12-21
Engineering Training Shapes	Require 3D models of different submarine engineering configurations to demonstrate mechanical sound sources	All LG in EO 002.02	LSR 10-21
Standardized Oceanography Briefing Template	Require a Fleet approved Standardized Oceanography Briefing template to be utilized by both trainees in class and fleet SONAR Ops	LG 005.02.04 PC 005	N/A

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		**Classroom**				**IT Assets (Trainees & Instructors)**					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
MODULE 1 – COMMON SONAR OP ACOUSTIC OPERATOR											
PO 002 – ANALYZE PASSIVE ACOUSTIC DATA											
LG 002.01.01	100	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.01.02	300	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.01.03	700	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
EC 002.01	150	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.01	50	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.02	300	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.03	100	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.04	250	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.05	100	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.06	250	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.07	200	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.08	500	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.09	1400	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
EC 002.02	250	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
LG 002.02.10	350	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
PC 002	250	ADAC Classroom	SECRET FVEY	1	13	0	0	13	0	1	Speakers & Headphones
PO 005 – BRIEF/DISSEMINATE INTELLIGENCE/OCEANOGRAPHIC INFORMATION											
LG 005.01.01	100	Classroom	UNCLASS	1	13	13	0	13	0	1	ARP system
LG 005.01.02	150	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system
LG 005.01.03	100	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system
LG 005.01.04	100	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system
LG 005.01.05	400	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system
EC 005.01	300	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system
LG 005.02.01	900	Classroom	UNCLASS	1	13	13	0	13	13	1	ARP system DWAN or GPNET
EC 005.02.01	300	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system DWAN or GPNET
LG 005.02.02	1200	Classroom	UNCLASS	1	13	13	0	13	13	1	DWAN or GPNET
LG 005.02.03	100	Classroom	UNCLASS	1	13	1	0	13	0	1	DWAN or GPNET
LG 005.02.04	750	Classroom	SECRET	1	13	0	1	13	0	1	CSNI
EC 005.02.02	400	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system DWAN or GPNET
PC 005	400	Classroom	UNCLASS	1	13	1	0	13	0	1	ARP system DWAN or GPNET
PO 004 – CONDUCT TARGET MOTION ANALYSIS (TMA)											
LG 004.02.01	500	Classroom	UNCLASS	1	0	0	0	0	0	1	
LG 004.02.02	150	Classroom	UNCLASS	1	0	0	0	0	0	1	
LG 004.02.03	100	Classroom	UNCLASS	1	13	0	0	0	0	1	
LG 004.02.04	50	Classroom	UNCLASS	1	13	0	0	0	0	1	

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
EC 004.02	100	Classroom	UNCLASS	1	13	0	0	0	0	1	
LG 004.01.01	600	SEACOT	SECRET	1	13	0	0	0	0	1	
EC 004.01	300	SEACOT	SECRET	1	13	0	0	0	0	1	
PC 004	300	SEACOT	SECRET	1	0	0	0	0	0	1	
PO 001 – CONDUCT PREPARATIONS FOR LAUNCHING EXPENDABLES											
LG 001.01.01	350	Classroom	UNCLASS	1	13	1	0	0	0	1	PTP & laptop w/WINRGA
LG 001.02.01	450	Classroom	UNCLASS	1	13	1	0	0	0	1	PTP & laptop w/WINRGA
PC 001	300	Classroom	UNCLASS	1	13	1	0	0	0	1	PTP & laptop w/WINRGA
PO 003 – OPERATE PASSIVE ACOUSTIC EQUIPMENT											
LG 003.01.01	500	Classroom	SECRET	1	13	0	0	13	0	1	SPS ADR Recorder
EC 003.01	300	Classroom	SECRET	1	13	0	0	13	0	1	SPS ADR Recorder
LG 003.02.01	1500	SEACOT	SECRET	1	0	0	0	0	0	1	
EC 003.02.01	300	SEACOT	SECRET	1	0	0	0	0	0	1	
LG 003.02.02	1000	SEACOT	SECRET	1	0	0	0	0	0	1	
EC 003.02.02	300	SEACOT	SECRET	1	0	0	0	0	0	1	
LG 003.02.03	1800	SEACOT	SECRET	1	0	0	0	0	0	1	
PC 003	300	SEACOT	SECRET	1	0	0	0	0	0	1	

ANNEX I – RECOMMENDED SCHEDULE

This schedule is reflective of the Course Duration table in Chapter 1. Purple periods represent BAT, TAT, & PT, orange periods represent ECs, and red periods represent PCs. Guest lecturers are highlighted. Each cell contains the PO/EO name in bold, the lesson/assessment number, and finally the lesson/assessment name.

MODULE 1 SCHEDULE

TIME	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Course Introduction & Administration (BAT)**	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**PT**	**PT**	**EC 002.01** – Perform Passive Aural Listening
Period 2 0850-0940	**Course Introduction & Administration (BAT)**	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EC 002.01** – Perform Passive Aural Listening
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**Course Introduction & Administration (TAT)**	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EC 002.01** – Perform Passive Aural Listening
Period 4 1040-1130	**ADAC In Routine (TAT)**	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**PT**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.01.01** – Determining blade/shaft configuration (Lecture)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.02.01** – Classify Weapons Systems (Lecture/EDI)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)
Period 6 1320-1410	**EO 002.01.01** – Determining blade/shaft configuration (Lecture)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)
Period 8 1510-1600	**EO 002.01.02** – Identifying aural sources (Lecture/EDI)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.01.03** – Perform Passive Aural Listing (EDI/Practical)	**EO 002.02.02** – Classify Weapons Systems Lab (Practical)	**EO 002.02.03** – Classifying Surface Vessels (Lecture/EDI)

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**EO 002.02.03** – Classifying Surface Vessels (Lecture/EDI)	**EO 002.02.05** – Classifying Diesel Submarines (Lecture/EDI)	**EO 002.02.07** – Classifying Nuclear Submarines (Lecture/EDI)	**PT**	**EO 002.02.09** – Consolidation Labs (Practical)
Period 2 0850-0940	**EO 002.02.04** – Classifying Surface Vessels Lab (Practical)	**EO 002.02.06** – Classifying Diesel Submarines lab (Practical)	**EO 002.02.07** – Classifying Nuclear Submarines (Lecture/EDI)	**PT**	**EO 002.02.09** – Consolidation Labs (Practical)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.02.04** – Classifying Surface Vessels Lab (Practical)	**EO 002.02.06** – Classifying Diesel Submarines lab (Practical)	**EO 002.02.07** – Classifying Nuclear Submarines (Lecture/EDI)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)
Period 4 1040-1130	**EO 002.02.04** – Classifying Surface Vessels Lab (Practical)	**EO 002.02.06** – Classifying Diesel Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.02.04** – Classifying Surface Vessels Lab (Practical)	**EO 002.02.06** – Classifying Diesel Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)
Period 6 1320-1410	**EO 002.02.04** – Classifying Surface Vessels Lab (Practical)	**EO 002.02.06** – Classifying Diesel Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.02.05** – Classifying Diesel Submarines (Lecture/EDI)	**EO 002.02.07** – Classifying Nuclear Submarines (Lecture/EDI)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)
Period 8 1510-1600	**PT**	**TAT**	**EO 002.02.08** – Classifying Nuclear Submarines lab (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**BAT**

TIME	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**EO 002.02.09** – Consolidation Labs (Practical)	**PT**	**PT**	**EC 002.02** – Perform basic passive acoustic analysis	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)
Period 2 0850-0940	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EC 002.02** – Perform basic passive acoustic analysis	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EC 002.02** – Perform basic passive acoustic analysis	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)
Period 4 1040-1130	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EC 002.02** – Perform basic passive acoustic analysis	**EC 002.02** – Perform basic passive acoustic analysis (Debrief)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)	**PC 002**
Period 6 1320-1410	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)	**PC 002**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.09** – Consolidation Labs (Practical)	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)	**PC 002**
Period 8 1510-1600	**PT**	**EO 002.02.09** – Consolidation Labs (Practical)	**TAT**	**EO 002.02.10** – Team Basic Passive Acoustic Analysis (Scenario)	**PC 002**

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**TAT**	**PT**	**PT**	**PT**	**EO 005.02.01** – Setup ARPS (EDI)
Period 2 0850-0940	**PC 002 (Debrief)**	**EO 005.01.03** – Effects of water masses on ASW Sensors (Lecture)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**PT**	**EO 005.02.01** – Setup ARPS (EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**BAT**	**EO 005.01.03** – Effects of water masses on ASW Sensors (Lecture)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)
Period 4 1040-1130	**EO 005.01.01** – Interpret intelligence/oceanographic data (Think/Share)	**EO 005.01.04** – Littoral ASW Ops (Lecture)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 005.01.01** – Interpret intelligence/oceanographic data (Think/Share)	**EO 005.01.04** – Littoral ASW Ops (Lecture)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)
Period 6 1320-1410	**EO 005.01.02** – Sound Propagation (Lecture/EDI)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)
1410-1420	**PC 002**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 005.01.02** – Sound Propagation (Lecture/EDI)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)
Period 8 1510-1600	**EO 005.01.02** – Sound Propagation (Lecture/EDI)	**EO 005.01.05** – FOM Calculations (Lecture/EDI)	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.01** – Interpret oceanographic data	**EO 005.02.01** – Setup ARPS (EDI)

TIME	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25
Period 1 0800-0850	**EO 005.02.01** – Setup ARPS (EDI)	**PT**	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**PT**	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
Period 2 0850-0940	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**PT**	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
Period 4 1040-1130	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
Period 6 1320-1410	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 005.02.01** – Setup ARPS (EDI)	**EC 005.02.01** – Setup ARPS	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)
Period 8 1510-1600	**EO 005.02.01** – Setup ARPS (EDI)	**TAT**	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**BAT**

TIME	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30
Period 1 0800-0850	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**PT**	**PT**	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Preparation)	**PC 005 (Preparation)**
Period 2 0850-0940	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**PT**	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Preparation)	**PC 005 (Preparation)**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 005.02.02** – Conduct Acoustic Range Predictions (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Preparation)	**PC 005 (Preparation)**
Period 4 1040-1130	**TAT**	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Preparation)	**PC 005 (Preparation)**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 005.02.03** – ASW Mission Brief (Lecture)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Presentation)	**PC 005 (Presentation)**
Period 6 1320-1410	**EO 005.02.03** – ASW Mission Brief (Lecture)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Presentation)	**PC 005 (Presentation)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Presentation)	**PC 005 (Presentation)**
Period 8 1510-1600	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EO 005.02.04** – Create Oceanographic Brief (Practical)	**EC 005.02.02** – Determine Oceanographic effects on Acoustic Ops (Presentation)	**PC 005 (Presentation)**

TIME	DAY 31	DAY 32	DAY 33	DAY 34	DAY 35
Period 1 0800-0850	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**PT**	**EO 004.02.03** – Range Calculations & RELVEL (Lecture/Demo)	**PT**	**EO 004.01.01** – Operate CMS for TMA (EDI)
Period 2 0850-0940	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.04** – Geographic Strip Plot (EDI)	**PT**	**EO 004.01.01** – Operate CMS for TMA (EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.02** – Validate TMA data	**EC 004.01** – Operate CMS for TMA
Period 4 1040-1130	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.02** – Validate TMA data	**EC 004.01** – Operate CMS for TMA
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.02** – Doppler Theory (Lecture/Table Top)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.01** – Operate CMS for TMA
Period 6 1320-1410	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.02** – Doppler Theory (Lecture/Table Top)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.01** – Operate CMS for TMA
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 004.02.01** – Line of Sound (Lecture/EDI)	**EO 004.02.02** – Doppler Theory (Lecture/Table Top)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.01** – Operate CMS for TMA
Period 8 1510-1600	**BAT**	**EO 004.02.03** – Range Calculations & RELVEL (Lecture/Demo)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EO 004.01.01** – Operate CMS for TMA (EDI)	**EC 004.01** – Operate CMS for TMA

TIME	DAY 36	DAY 37			
Period 1 0800-0850	**PT**	**PT**			
Period 2 0850-0940	**PC 004**	**TAT (MOD 1 Complete)**			
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**PC 004**	**TAT (MOD 1 Complete)**			
Period 4 1040-1130	**PC 004**	**BAT (MOD 1 Complete)**			
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**PC 004**	**BAT (MOD 1 Complete)**			
Period 6 1320-1410	**PC 004**	**BAT (MOD 1 Complete)**			
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PC 004**	**BAT (MOD 1 Complete)**			
Period 8 1510-1600	**TAT (Mod 1 Complete)**				

MODULE 2 SCHEDULE

TIME	DAY 1	DAY 2	DAY 38	DAY 4	DAY 5
Period 1 0800-0850	**BAT**	**PT**	**EO 001.02.01** – Upload to EMATT (EDI)	**PT**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)
Period 2 0850-0940	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PT**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)
Period 4 1040-1130	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.02.01** – Operate SPS (EDI)
Period 6 1320-1410	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.02.01** – Operate SPS (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.02.01** – Operate SPS (EDI)
Period 8 1510-1600	**EO 001.01.01** – Conduct Preparations for Launching Expendables (Lecture/EDI)	**EO 001.02.01** – Upload to EMATT (EDI)	**PC 001**	**EO 003.01.01** – Operate SPS Recorder (Lecture/EDI)	**EO 003.02.01** – Operate SPS (EDI)

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**PT**	**EO 003.02.01** – Operate SPS (EDI)
Period 2 0850-0940	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)
Period 4 1040-1130	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)
Period 6 1320-1410	**EC 003.01** – Record acoustic intelligence data	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PT**	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)
Period 8 1510-1600	**PT**	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.01** – Operate SPS (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)

TIME	DAY 11		DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**PT**	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
Period 2 0850-0940	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
Period 4 1040-1130	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
Period 6 1320-1410	**EC 003.02.01** – Operate SPS	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EC 003.02.02** – Operate SPS for Anti-Torpedo procedures	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**PT**	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission
Period 8 1510-1600	**PT**	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.02** – Operate SPS for Anti-Torpedo procedures (EDI)	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**PT**	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PT**	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**
Period 2 0850-0940	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PT**
Period 4 1040-1130	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**TAT (Grad Parade)**
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**	**TAT (Course Reports)**
Period 6 1320-1410	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**	**BAT (Out Routine)**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**	
Period 8 1510-1600	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**EO 003.02.03** – Consolidation - Operate SPS for ASW Mission	**PC 003**	